

T3 – CHARACTERIZING AMBIGUITIES IN SPARSE ARRAYS

IEEE SAM Workshop

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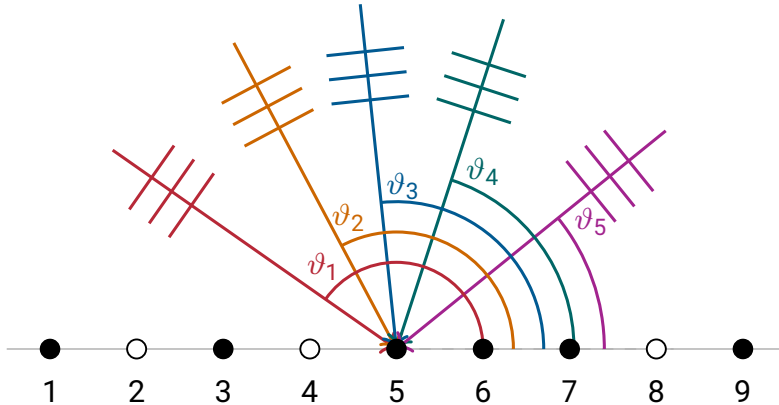
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Open6GHub+

SPARSE ARRAYS

Thinned arrays



SPARSE ARRAYS

Sparse arrays offer several benefits:

- Large aperture
- Low hardware cost/ complexity

How to optimally design sparse arrays?

- Optimized **beampattern** (Mainlobe width vs sidelobe/grating level)
- Optimized **performance bounds** (CRB,...)^{[1][2]}
- Optimized co-array properties^{[3][4][5]}

Problem with these designs:

- Beampattern and Performance bounds do not capture the essential aspects of the complex-valued **interplay between multiple-sources**.
- Performance bounds are often **local**.

[1] J. Tabrikian et al., "Cognitive Antenna Selection for Automotive Radar Using Bobrovsky-Zakai Bound", 2021

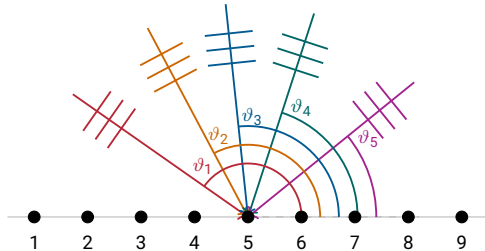
[2] I. van der Werf et al., "Jointly Optimal Array Geometries and Waveforms in Active Sensing: New Insights Into Array Design via the Cramér-Rao Bound", 2025

[3] P. Pal and P. P. Vaidyanathan, "Nested Arrays: A Novel Approach to Array Processing With Enhanced Degrees of Freedom", 2010

[4] P. P. Vaidyanathan and P. Pal, "Sparse Sensing With Co-Prime Samplers and Arrays", 2011

[5] X. Wang et al., "Analysis and Design of Optimum Sparse Array Configurations for Adaptive Beamforming", 2018

DOA ESTIMATION IN UNIFORM LINEAR ARRAYS

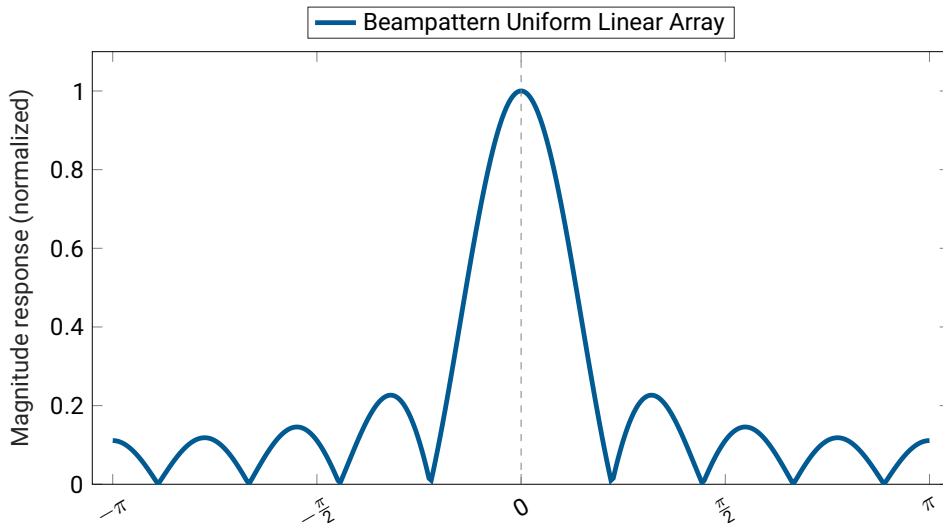


$$\mathbf{A}(z_1, z_2, \dots, z_5) = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ z_1 & z_2 & z_3 & z_4 & z_5 \\ z_1^2 & z_2^2 & z_3^2 & z_4^2 & z_5^2 \\ z_1^3 & z_2^3 & z_3^3 & z_4^3 & z_5^3 \\ z_1^4 & z_2^4 & z_3^4 & z_4^4 & z_5^4 \\ z_1^5 & z_2^5 & z_3^5 & z_4^5 & z_5^5 \\ z_1^6 & z_2^6 & z_3^6 & z_4^6 & z_5^6 \\ z_1^7 & z_2^7 & z_3^7 & z_4^7 & z_5^7 \\ z_1^8 & z_2^8 & z_3^8 & z_4^8 & z_5^8 \end{bmatrix}$$

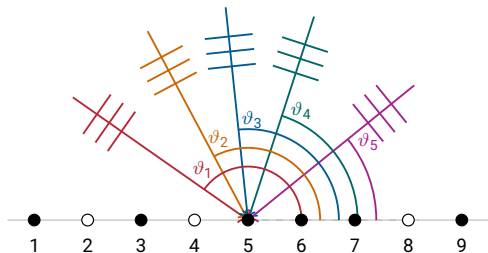
$$\mathbf{y} = \mathbf{A}(z_1, \dots, z_L) \mathbf{x} + \text{noise}$$

- $\mathbf{y} = [y_1, \dots, y_M]^T$ are the sensor measurements
- $\mathbf{x} = [x_1, \dots, x_L]^T$ is the signal waveform
- $\mathbf{A}(z_1, \dots, z_L) = [\mathbf{a}(z_1), \dots, \mathbf{a}(z_L)]$ is a full $M \times L$ **Vandermonde matrix**
- Generators $z_\ell = e^{j2\pi\theta_\ell}$
- $\theta_\ell = \frac{2\pi d}{\lambda} \cos \vartheta_\ell$ are the **electrical angles** corresponding to ϑ_ℓ .

GRATING LOBES IN SPARSE ARRAYS



DOA ESTIMATION IN THINNED UNIFORM LINEAR ARRAYS



$$\mathbf{A}_{\mathcal{I}}(z_1, z_2, \dots, z_5) =$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ \cancel{z_1} & \cancel{z_2} & \cancel{z_3} & \cancel{z_4} & \cancel{z_5} \\ z_1^2 & z_2^2 & z_3^2 & z_4^2 & z_5^2 \\ \cancel{z_1^3} & \cancel{z_2^3} & \cancel{z_3^3} & \cancel{z_4^3} & \cancel{z_5^3} \\ z_1^4 & z_2^4 & z_3^4 & z_4^4 & z_5^4 \\ z_1^5 & z_2^5 & z_3^5 & z_4^5 & z_5^5 \\ z_1^6 & z_2^6 & z_3^6 & z_4^6 & z_5^6 \\ \cancel{z_1^7} & \cancel{z_2^7} & \cancel{z_3^7} & \cancel{z_4^7} & \cancel{z_5^7} \\ \cancel{z_1^8} & \cancel{z_2^8} & \cancel{z_3^8} & \cancel{z_4^8} & \cancel{z_5^8} \end{bmatrix}$$

$$\mathbf{y}_{\mathcal{I}} = \mathbf{A}_{\mathcal{I}}(z_1, \dots, z_L) \mathbf{x} + \text{noise}$$

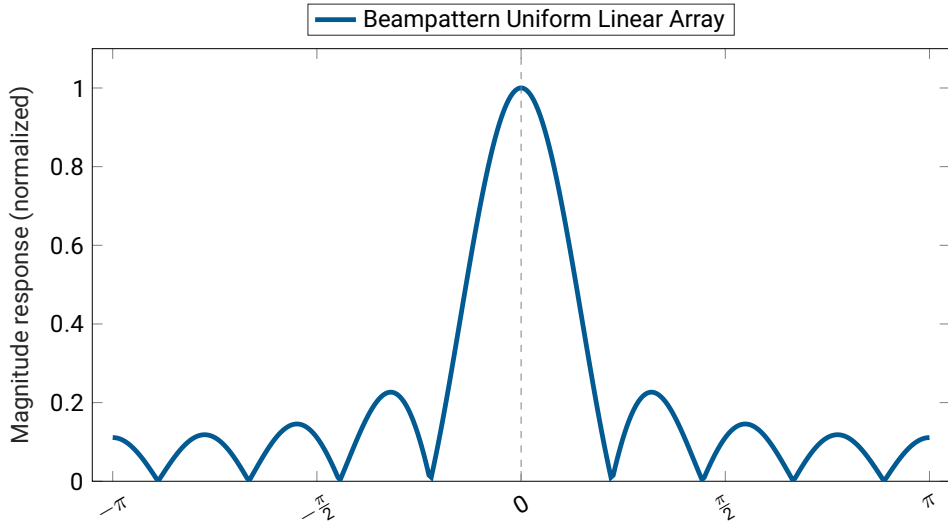
- Index set of sensors kept:

$$\mathcal{I} = \{1, 3, 5, 6, 7, 9\},$$

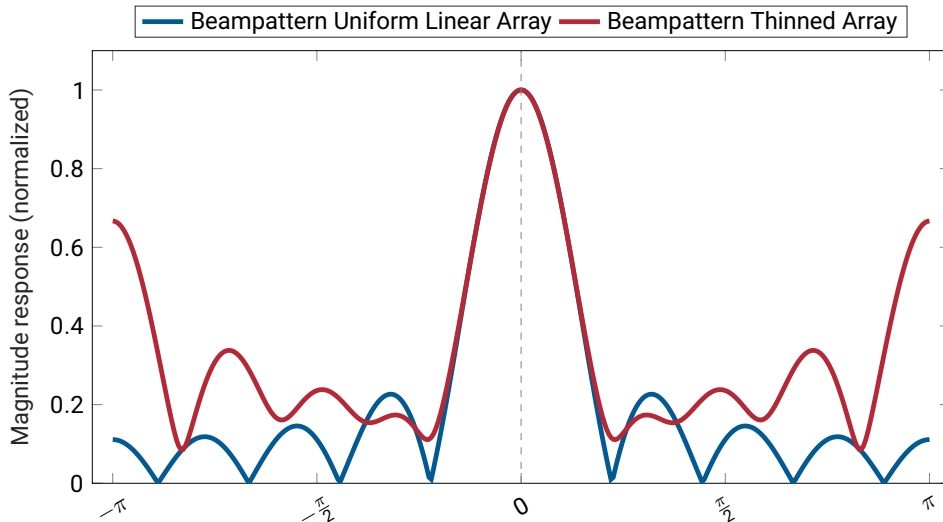
- Complementary index set of removed sensors: complementary set

$$\bar{\mathcal{I}} = \{2, 4, 8\},$$

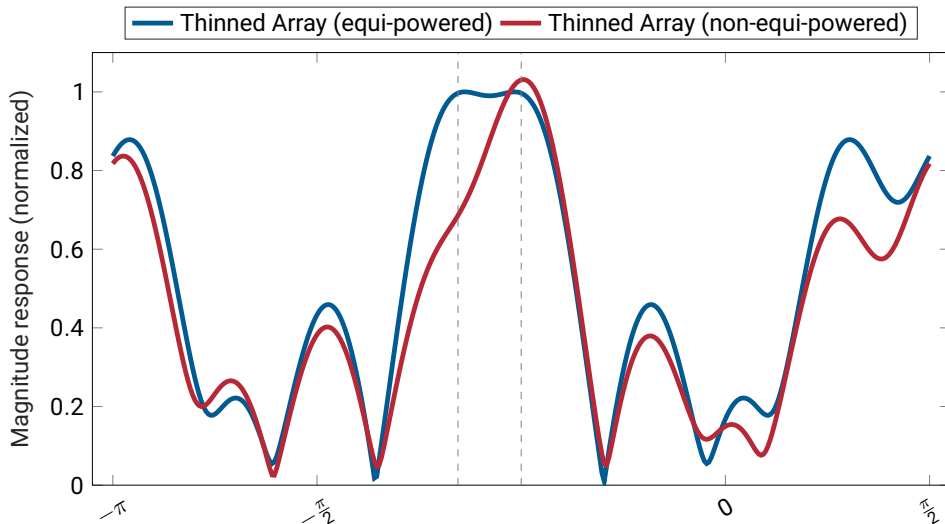
GRATING LOBES IN SPARSE ARRAYS



GRATING LOBES IN SPARSE ARRAYS

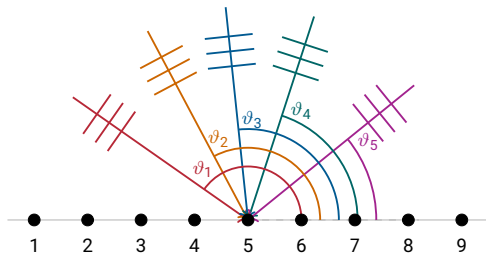


GRATING LOBES IN SPARSE ARRAYS





RANK PROPERTY OF STEERING MATRIX AND UNIQUENESS



$$\mathbf{A}(z_1, z_2, \dots, z_5) = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ z_1 & z_2 & z_3 & z_4 & z_5 \\ z_1^2 & z_2^2 & z_3^2 & z_4^2 & z_5^2 \\ z_1^3 & z_2^3 & z_3^3 & z_4^3 & z_5^3 \\ z_1^4 & z_2^4 & z_3^4 & z_4^4 & z_5^4 \\ z_1^5 & z_2^5 & z_3^5 & z_4^5 & z_5^5 \\ z_1^6 & z_2^6 & z_3^6 & z_4^6 & z_5^6 \\ z_1^7 & z_2^7 & z_3^7 & z_4^7 & z_5^7 \\ z_1^8 & z_2^8 & z_3^8 & z_4^8 & z_5^8 \end{bmatrix}$$

$$\mathbf{y} = \mathbf{A}(z_1, \dots, z_L) \mathbf{x} + \text{noise}$$

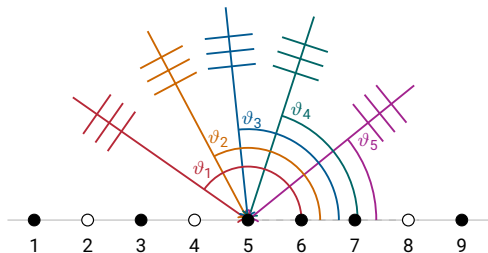
Full Vandermonde matrix $\mathbf{A}(z_1, z_2, z_3, \dots, z_L)$ is always full rank.



Parameter estimates are **unique**.^[6]

[6] M. Wax and I. Ziskind, "On unique localization of multiple sources by passive sensor arrays", Jul. 1989

RANK PROPERTY OF STEERING MATRIX AND UNIQUENESS



$$\mathbf{A}_{\mathcal{I}}(z_1, z_2, \dots, z_5) =$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ \cancel{z_1^1} & \cancel{z_2^1} & \cancel{z_3^1} & \cancel{z_4^1} & \cancel{z_5^1} \\ z_1^2 & z_2^2 & z_3^2 & z_4^2 & z_5^2 \\ \cancel{z_1^3} & \cancel{z_2^3} & \cancel{z_3^3} & \cancel{z_4^3} & \cancel{z_5^3} \\ \cancel{z_1^4} & \cancel{z_2^4} & \cancel{z_3^4} & \cancel{z_4^4} & \cancel{z_5^4} \\ z_1^5 & z_2^5 & z_3^5 & z_4^5 & z_5^5 \\ \cancel{z_1^6} & \cancel{z_2^6} & \cancel{z_3^6} & \cancel{z_4^6} & \cancel{z_5^6} \\ \cancel{z_1^7} & \cancel{z_2^7} & \cancel{z_3^7} & \cancel{z_4^7} & \cancel{z_5^7} \\ \cancel{z_1^8} & \cancel{z_2^8} & \cancel{z_3^8} & \cancel{z_4^8} & \cancel{z_5^8} \end{bmatrix}$$

- **Reversely**, if the thinned steering matrix $\mathbf{A}_{\mathcal{I}}(z_1, z_2, z_3, \dots, z_5)$ drops rank $\Leftrightarrow$ there exist two different parameter sets $\{z_1, z_2, \mathbf{x}_1\}$ and $\{z_3, z_4, z_5, \mathbf{x}_2\}$ such that:^{[7][8][9]}

$$\mathbf{y}_{\mathcal{I}} = \mathbf{A}_{\mathcal{I}}(z_1, z_2) \mathbf{x}_1 = -\mathbf{A}_{\mathcal{I}}(z_3, z_4, z_5) \mathbf{x}_2$$

[7] A. Manikas and C. Proukakis, "Modeling and estimation of ambiguities in linear arrays", Aug. 1998

[8] Y. Abramovich et al., "Identifiability and manifold ambiguity in DOA estimation for nonuniform linear antenna arrays", Mar. 1999

[9] B. Friedlander, "Multi-Signal Ambiguity in Direction-of-Arrival Estimation", 2023

BEAMPATTERN FOR AMBIGUOUS ARRAY

Example 6 of <http://arxiv.org/abs/2606.00360>

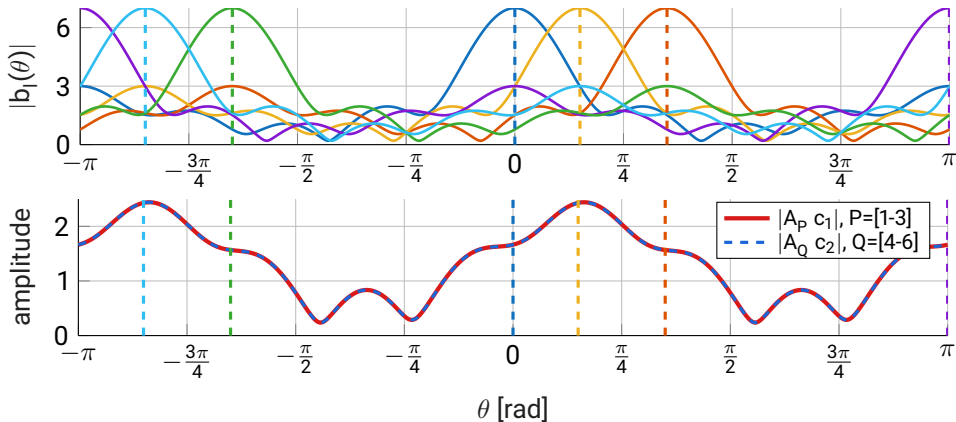


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Introduction

Rank conditions and identifiability

SINGLE SNAPSHOT / COHERENT SOURCE CASE

How is the number of uniquely identifiable sources related to the Kruskal rank of the steering matrix?

- Let K_{rank} denote the **Kruskal rank** of the steering matrix $\mathbf{A}_{\mathcal{I}}(z_1, \dots, z_{K_{\text{rank}}})$, i.e., the **largest number of columns** with distinct generators, for which the steering matrix is always **full column rank**.
- Nullspace property:** There exists $K_{\text{rank}} + 1$ distinct generators $z_1, \dots, z_{K_{\text{rank}}}, z_{K_{\text{rank}}+1}$ and linear coefficients $x_1, \dots, x_{K_{\text{rank}}}, x_{K_{\text{rank}}+1}$ such that

$$[\mathbf{A}(z_1, \dots, z_{K_{\text{rank}}}), \mathbf{a}(z_{K_{\text{rank}}+1})] \begin{bmatrix} x_1 \\ \vdots \\ x_{K_{\text{rank}}} \\ x_{K_{\text{rank}}+1} \end{bmatrix} = \mathbf{0}$$

- Spark:** The spark $(\mathbf{A}) = K_{\text{rank}} + 1$ is defined as **the smallest number of columns** of an arbitrary steering matrix $\mathbf{A}$ with distinct generators, that are **linearly dependent**.

SINGLE SNAPSHOT / COHERENT SOURCE CASE

How is the number of uniquely identifiable sources related to the Kruskal rank of the steering matrix?

- Hence for $L' = \lceil \frac{K_{\text{rank}}+1}{2} \rceil$ the set of generators can be partitioned as $z_1, \dots, z_{L'}, z_{L'+1}, \dots, z_{K_{\text{rank}}+1}$ such that

$$\mathbf{y} = \mathbf{A}(z_1, \dots, z_{L'}) \begin{bmatrix} x_1 \\ \vdots \\ x_{L'} \end{bmatrix} = -\mathbf{A}(z_{L'+1}, \dots, z_{K_{\text{rank}}+1}) \begin{bmatrix} x_{L'} \\ \vdots \\ x_{K_{\text{rank}}+1} \end{bmatrix}$$

- In other words, the **measurements** $\mathbf{y}$ can **indistinguishably** be explained by the two **distinct parameter sets**

$$\{z_1, \dots, z_{L'}, x_1, \dots, x_{L'}\} \neq \{z_{L'+1}, \dots, z_{K_{\text{rank}}+1}, x_{L'}, \dots, x_{K_{\text{rank}}+1}\}$$

- Thus, in the **single snapshot / coherent source case**, the number of uniquely identifiable sources^[6]

$$L' < \left\lceil \frac{K_{\text{rank}} + 1}{2} \right\rceil.$$

[6] M. Wax and I. Ziskind, "On unique localization of multiple sources by passive sensor arrays", Jul. 1989



η -NON-COHERENT WAVEFORM VECTORS / SOURCES

How is the number of uniquely identifiable sources related to the Kruskal rank of the steering matrix?

- With K_{rank} denoting the **Kruskal rank** of the steering matrix $\mathbf{A}_{\mathcal{I}}(z_1, \dots, z_{K_{\text{rank}}})$.
- Nullspace property:** There exist generators $z_1, \dots, z_{K_{\text{rank}}}, z_{K_{\text{rank}}+1}, \dots, z_{K_{\text{rank}}+\eta}$ and linear coefficients $x_1(t_i), \dots, x_{K_{\text{rank}}}(t_i), x_{K_{\text{rank}}+1}(t_i), \dots, x_{K_{\text{rank}}+\eta}(t_i)$ for $\eta \leq K_{\text{rank}}$ and $t_i \in \{t_1, \dots, t_\eta\}$ such that

$$\underbrace{\left[\underbrace{\mathbf{A}(z_1, \dots, z_{K_{\text{rank}}})}_{\text{linearly independent}}, \mathbf{a}(z_{K_{\text{rank}}+1}), \dots, \mathbf{a}(z_{K_{\text{rank}}+\eta}) \right]}_{\text{Nullspace dimension}=\eta} \underbrace{\begin{bmatrix} x_1(t_1) & \cdots & x_1(t_\eta) \\ \vdots & \vdots & \vdots \\ x_{K_{\text{rank}}}(t_1) & \cdots & x_{K_{\text{rank}}}(t_\eta) \\ x_{K_{\text{rank}}+1}(t_1) & \cdots & x_{K_{\text{rank}}+1}(t_\eta) \\ \vdots & \vdots & \vdots \\ x_{K_{\text{rank}}+\eta}(t_1) & \cdots & x_{K_{\text{rank}}+\eta}(t_\eta) \end{bmatrix}}_{\eta \text{ linearly independent waveform vectors}} = \mathbf{0}$$

η -NON-COHERENT WAVEFORM VECTORS / SOURCES

How is the number of uniquely identifiable sources related to the Kruskal rank of the steering matrix?

- As before, now for $L' = \lceil \frac{K_{\text{rank}} + \eta}{2} \rceil$ with $\eta \leq K_{\text{rank}}$ the set of generators can be partitioned as $\mathbf{z}_1, \dots, \mathbf{z}_{L'}, \mathbf{z}_{L'+1}, \dots, \mathbf{z}_{K_{\text{rank}} + \eta}$ such that

$$\mathbf{y} = \mathbf{A}(\mathbf{z}_1, \dots, \mathbf{z}_{L'}) \begin{bmatrix} x_1(t_1) & \cdots & x_1(t_\eta) \\ \vdots & \vdots & \vdots \\ x_{L'}(t_1) & \cdots & x_{L'}(t_\eta) \end{bmatrix} = -\mathbf{A}(\mathbf{z}_{L'+1}, \dots, \mathbf{z}_{K_{\text{rank}} + \eta}) \begin{bmatrix} x_{L'}(t_1) & \cdots & x_{L'}(t_\eta) \\ \vdots & \vdots & \vdots \\ x_{K_{\text{rank}} + \eta}(t_1) & \cdots & x_{K_{\text{rank}} + \eta}(t_\eta) \end{bmatrix}$$

- Hence, the measurements $\mathbf{y}$ are **indistinguishable** under the two **distinct parameter sets**

$$\{\mathbf{z}_1, \dots, \mathbf{z}_{L'}, x_1(t_1), \dots, x_{L'}(t_\eta)\} \neq \{\mathbf{z}_{L'+1}, \dots, \mathbf{z}_{K_{\text{rank}} + \eta}, x_{L'}(t_i), \dots, x_{K_{\text{rank}} + \eta}(t_i)\}$$

- Thus, in the η non-coherent waveform / snapshot / source case, the number of **uniquely identifiable** sources^[6]

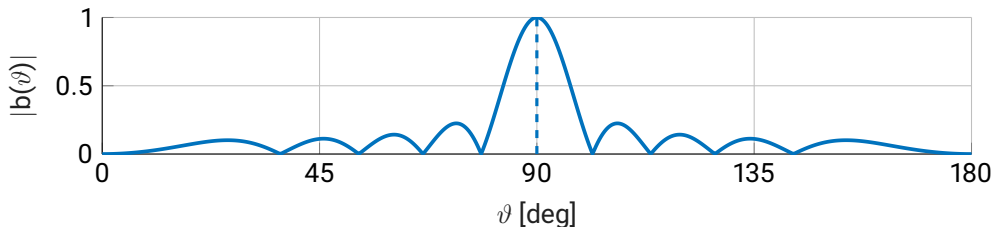
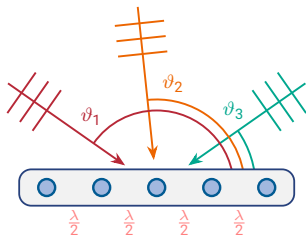
$$L' < \left\lceil \frac{K_{\text{rank}} + \eta}{2} \right\rceil.$$

[6] M. Wax and I. Ziskind, "On unique localization of multiple sources by passive sensor arrays", Jul. 1989

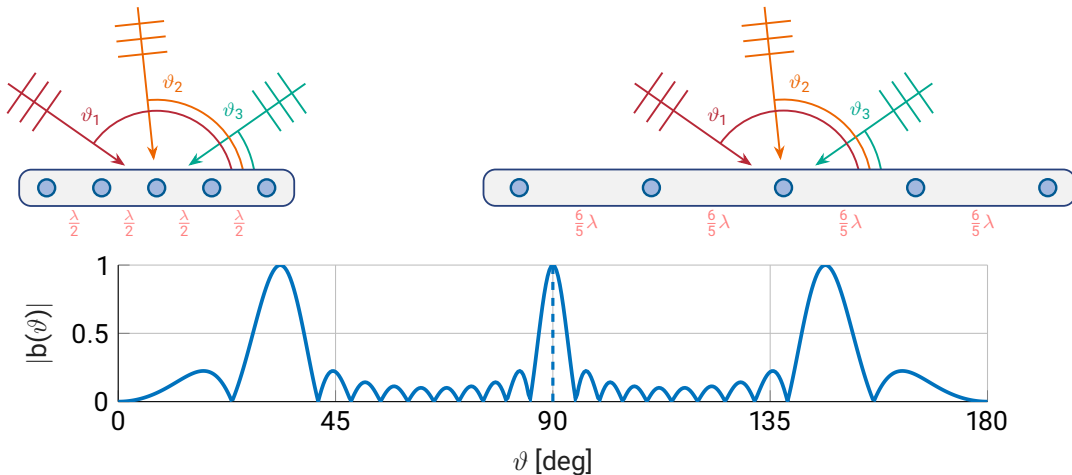
Introduction

Array ambiguities

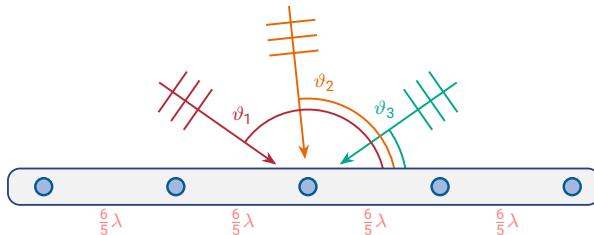
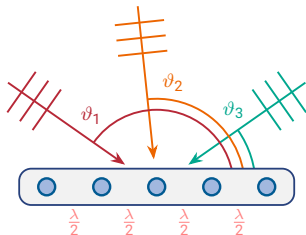
FIRST-ORDER AMBIGUITIES: GRATING LOBES



FIRST-ORDER AMBIGUITIES: SPATIAL SAMPLING THEOREM



FIRST-ORDER AMBIGUITIES: GRATING LOBES



- Array steering vectors for different directions become identical, i.e., for $d = \frac{6}{5}\lambda$

$$\mathbf{a}(\vartheta) = \begin{bmatrix} 1 \\ e^{j\frac{2\pi d}{\lambda} \cos \vartheta_\ell} \\ e^{j\frac{4\pi d}{\lambda} \cos \vartheta_\ell} \\ \vdots \\ e^{j\frac{2\pi(M-1)d}{\lambda} \cos \vartheta_\ell} \end{bmatrix} = \begin{bmatrix} 1 \\ e^{j2\pi\frac{6}{5} \cos \vartheta_\ell} \\ e^{j2\pi2\frac{6}{5} \cos \vartheta_\ell} \\ \vdots \\ e^{j2\pi(M-1)\frac{6}{5} \cos \vartheta_\ell} \end{bmatrix} \Rightarrow \mathbf{a}(\vartheta) \big|_{\vartheta=90^\circ} = \mathbf{a}(\vartheta) \big|_{\vartheta=\arccos \frac{5}{6} \approx 33.56^\circ \wedge 146.44^\circ} = \mathbf{1}$$

FIRST-ORDER AMBIGUITIES: GRATING LOBES

- For **first-order ambiguities** with $\mathbf{a}(\vartheta') = \mathbf{a}(\vartheta'')$ the corresponding array steering matrix

$$\mathbf{A}(\vartheta', \vartheta'', \vartheta_3, \dots, \vartheta_L) = [\mathbf{a}(\vartheta'), \mathbf{a}(\vartheta''), \mathbf{a}(\vartheta_3), \dots, \mathbf{a}(\vartheta_L)]$$

has two (or multiple) **identical columns**, hence the matrix is rank deficient.

- First-order ambiguities are **trivial** and comparably **easy to detect and handle**, e.g., by limiting the field of view.
- In practice, first-order ambiguities may also be suppressed by the superimposed element pattern.

HIGHER-ORDER AMBIGUITIES

- Higher-order ambiguities emerge in the case of non-trivial rank drop conditions of the array steering matrix

$$\mathbf{A}_{\mathcal{I}}(\vartheta_1, \vartheta_2, \dots, \vartheta_L) = [\mathbf{a}_{\mathcal{I}}(\vartheta_1), \mathbf{a}_{\mathcal{I}}(\vartheta_2), \dots, \mathbf{a}_{\mathcal{I}}(\vartheta_L)],$$

i.e., when $L \leq M_{\mathcal{I}}$ distinct steering vectors $\mathbf{a}_{\mathcal{I}}(\vartheta_1) \neq \mathbf{a}_{\mathcal{I}}(\vartheta_2) \neq \dots \neq \mathbf{a}_{\mathcal{I}}(\vartheta_L)$ become linearly dependent.

- In this case there exists a linear combination with coefficients $\alpha_1, \alpha_2, \dots, \alpha_L$ such that $\sum_{\ell=1}^L \alpha_{\ell} \mathbf{a}_{\mathcal{I}}(\vartheta_{\ell}) = \mathbf{0}$, hence the vector $\alpha = [\alpha_1, \alpha_2, \dots, \alpha_L]^T$ lies in the nullspace of $\mathbf{A}_{\mathcal{I}}(\vartheta_1, \vartheta_2, \dots, \vartheta_L)$, i.e., the nullspace

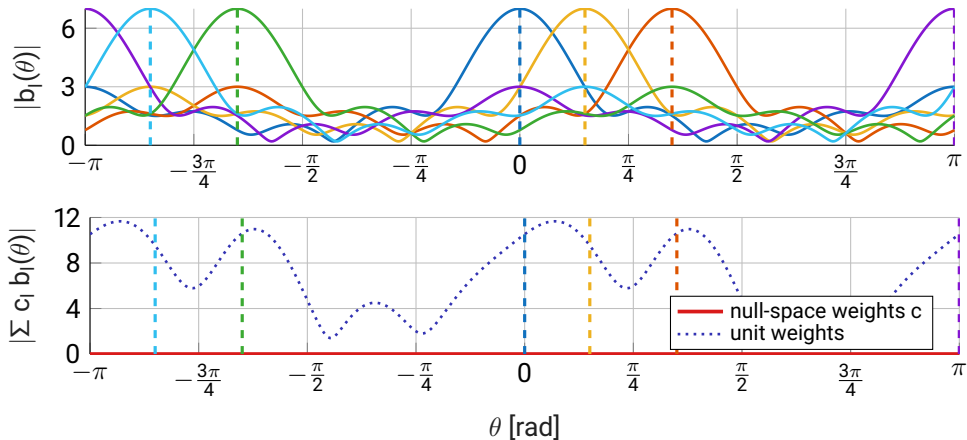
$$\mathcal{N}(\mathbf{A}_{\mathcal{I}}(\vartheta_1, \vartheta_2, \dots, \vartheta_L)) = \{\gamma \in \mathbb{C}^L \mid \mathbf{A}_{\mathcal{I}}(\vartheta_1, \vartheta_2, \dots, \vartheta_L) \gamma = \mathbf{0}\}$$

is non-empty and

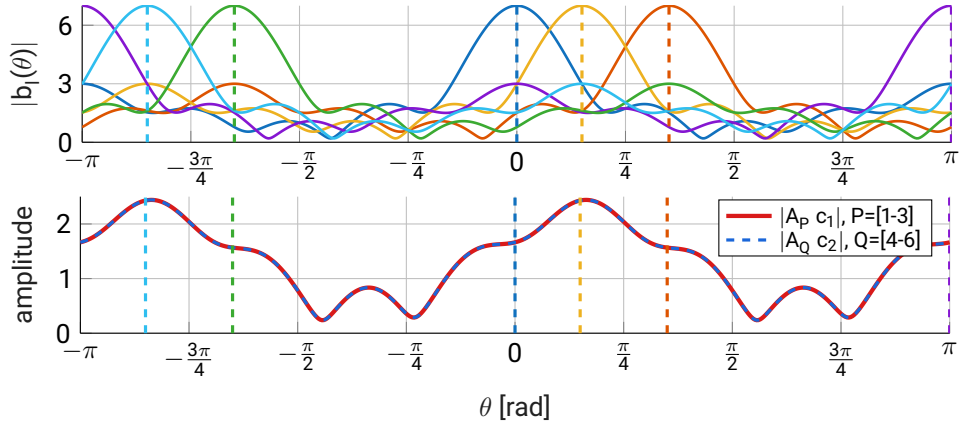
$$\alpha \in \mathcal{N}(\mathbf{A}_{\mathcal{I}}(\vartheta_1, \vartheta_2, \dots, \vartheta_L)).$$

- Determining the set of all higher-order ambiguities for arbitrary arrays or for sparse (thinned) linear arrays is non-trivial. In the following we focus on thinned ULAs.

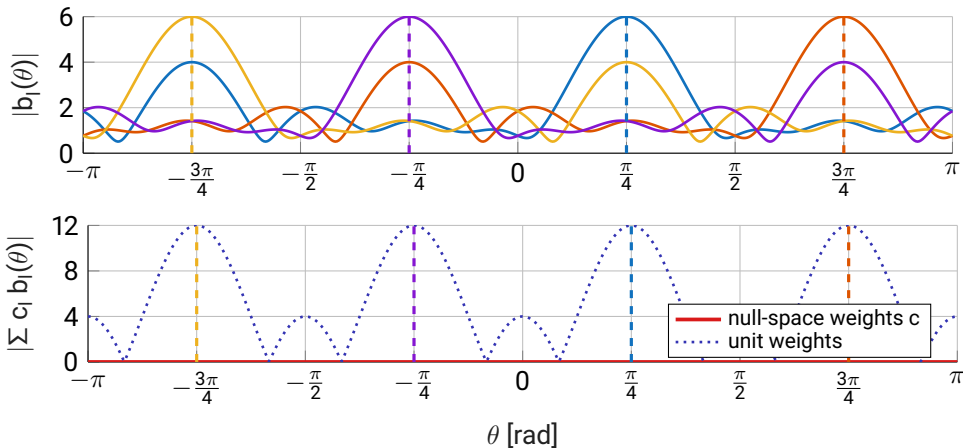
BEAMPATTERNS: EXAMPLE 6



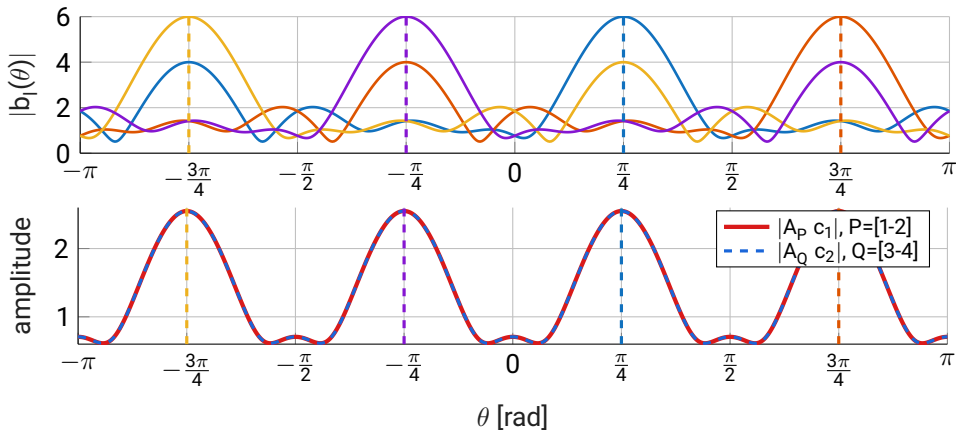
BEAMPATTERNS: EXAMPLE 6



BEAMPATTERNS: EXAMPLE 7



BEAMPATTERNS: EXAMPLE 7



Sparse array analysis

State of the art

SPARSE ARRAYS: THINNED ULA

- **Uniform Linear Array (ULA):** Sensors on a regular grid with equal inter-sensor spacing
- **Sparse arrays:** Sensors are removed, leaving **irregular gaps** in the grid

 $\mathbf{a}_{\text{ULA}}$

$$\begin{bmatrix} 1 & z^1 & z^2 & z^3 & z^4 \end{bmatrix}^T$$

 thinned

 $\mathbf{a}_{\text{sparse}}$

$$\begin{bmatrix} 1 & z^1 & \square & \square & z^4 \end{bmatrix}^T$$

SPARSE ARRAYS: UNIFORM AMBIGUITIES

Existing results



- Manikas and Proukakakis identified a set of **uniformly** (in electrical angle) **spaced ambiguity** generators, for which the steering matrix drops rank.
- Given the sensor locations corresponding to the index set

$$\mathcal{I} = \{m_1, m_2, \dots, m_{M_{\mathcal{I}}}\}, \quad 1 = m_1 < m_2 < \dots < m_{M_{\mathcal{I}}} = M$$

ambiguities emerge if a generator set

$$\left\{ 1, e^{j\pi \frac{2}{|m_i - m_j|}}, e^{j\pi \frac{4}{|m_i - m_j|}}, \dots, e^{j\pi \frac{2c}{|m_i - m_j|}} \right\}$$

exists, where $i \neq j \in [1, \dots, M]$ and $c \in \mathbb{N}$ is the largest integer satisfying $c < |m_i - m_j|$ and if the set contains at least M elements, i.e., $c \geq (M - 1)$. [7]

[7] A. Manikas and C. Proukakakis, "Modeling and estimation of ambiguities in linear arrays", Aug. 1998

SPARSE ARRAYS: UNIFORM AMBIGUITIES

Existing results

- To see that M generators from the set

$$\left\{ 1, e^{j\pi \frac{2}{|m_i - m_j|}}, e^{j\pi \frac{4}{|m_i - m_j|}}, \dots, e^{j\pi \frac{2c}{|m_i - m_j|}} \right\}$$

lead to a rank drop in the array steering matrix $\mathbf{A}$, we choose without loss of generality m_i as the origin, i.e., $m_i = 0$.

- Then the m_i -th and the m_j -th row of the steering matrix are given by

$$[1, e^{j0\pi \frac{2}{|m_j|}}, e^{j0\pi \frac{4}{|m_j|}}, \dots, e^{j0\pi \frac{2c}{|m_j|}}]^T = [1, 1, 1, \dots, 1]^T$$

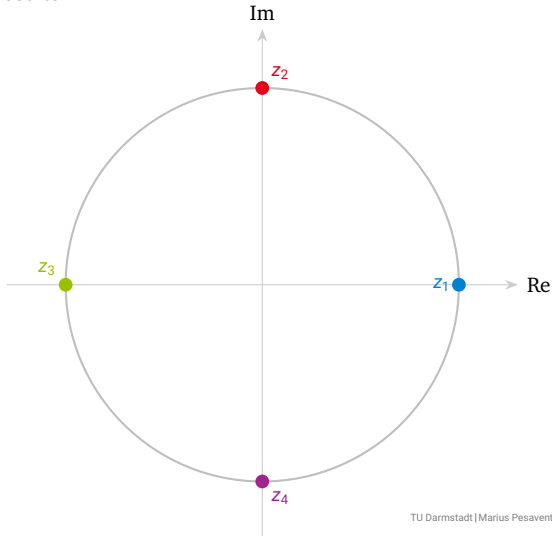
and

$$[1, e^{jm_j\pi \frac{2}{|m_j|}}, e^{jm_j\pi \frac{4}{|m_j|}}, \dots, e^{jm_j\pi \frac{2c}{|m_j|}}]^T = [1, 1, 1, \dots, 1]^T$$

respectively. Hence, the rows become identical and the steering matrix drops rank.

UNIFORM AMBIGUITIES

Existing results



Uniform generator set:

$$\arg z_1 = 0, \quad \arg z_2 = \frac{\pi}{2},$$

$$\arg z_3 = \pi, \quad \arg z_4 = \frac{3\pi}{2},$$

Any 3 columns of matrix

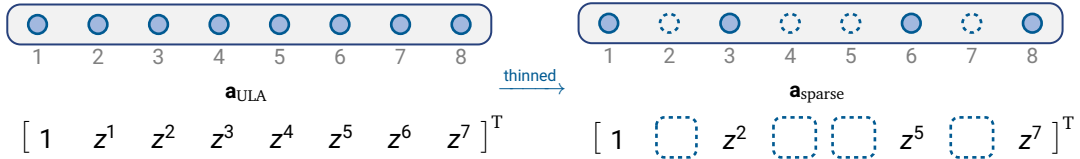
$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & e^{j\frac{\pi}{2}} & e^{j\pi} & e^{j\frac{3\pi}{2}} \\ 1 & e^{j4\frac{\pi}{2}} & e^{j4\pi} & e^{j4\frac{3\pi}{2}} \end{bmatrix}$$

are linearly dependent.

SPARSE SYMMETRIC ARRAY: THINNED ULA

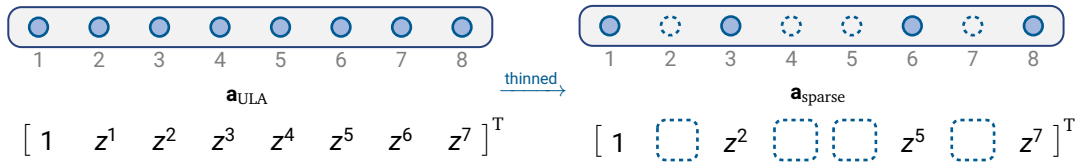
Existing results

- **Uniform Linear Array (ULA):** Sensors on a regular grid with equal inter-sensor spacing
- **Sparse arrays:** Sensors are removed at symmetric positions around the center, leaving **irregular gaps** in the grid



NON-UNIFORM AMBIGUITIES

Existing results



- If the (square) sparse array steering matrix

$$\mathbf{A}_{\mathcal{I}}(z_1, \dots, z_4) = \begin{bmatrix} 1 & 1 & 1 & 1 \\ z_1^2 & z_2^2 & z_3^2 & z_4^2 \\ z_1^5 & z_2^5 & z_3^5 & z_4^5 \\ z_1^7 & z_2^7 & z_3^7 & z_4^7 \end{bmatrix}$$

drops rank, then there must exist a non-zero nullspace vector $\mathbf{p}$ such that $\mathbf{A}_{\mathcal{I}}^T \mathbf{p} = \mathbf{0}$, i.e., $\mathbf{p} \in \mathcal{N}(\mathbf{A}_{\mathcal{I}}^T(z_1, \dots, z_4))$.

NON-UNIFORM AMBIGUITIES

Existing results

- Manikas and Proukakis^[7] and later Dowlut^[11] **reversed** this **argument**. They constructed particular vectors $\{\mathbf{p}_1, \dots, \mathbf{p}_{M_{\mathcal{I}}}\}$, motivated by differential geometry, and tested if corresponding structure nullspace matrices $\mathbf{A}_{\mathcal{I}}(z_{r,1}, \dots, z_{r,L})$ exist, for which

$$\mathbf{A}_{\mathcal{I}}^T(z_{r,1}, \dots, z_{r,L})\mathbf{p}_r = \mathbf{0}, \quad r = 1, \dots, M_{\mathcal{I}}$$

- This conditions essentially requires that the **symmetric sparse steering vectors** $\mathbf{a}_{\mathcal{I}}(z)$ for all $z \in \{z_{r,1}, \dots, z_{r,L}\}$ satisfy

$$P_r(z) = \mathbf{a}_{\mathcal{I}}^T(z)\mathbf{p}_r = \sum_{\ell=0}^{L-1} p_{r,\ell+1}z^\ell = 0.$$

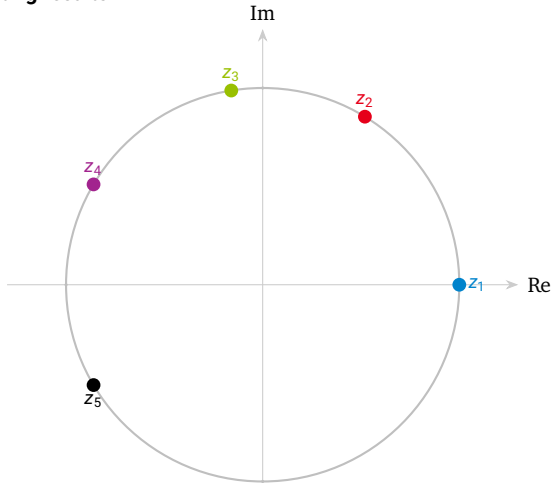
- Hence, the **roots** of $P_r(z)$, denoted by $\{z_{r,1}, \dots, z_{r,L}\}$ form a **generator set** for which the corresponding steering matrix $\mathbf{A}_{\mathcal{I}}(z_{r,1}, \dots, z_{r,L})$ becomes **rank deficient**.

[7] A. Manikas and C. Proukakis, "Modeling and estimation of ambiguities in linear arrays", Aug. 1998

[11] N. Dowlut, "An extended ambiguity criterion for array design", 2002

NON-UNIFORM AMBIGUITY SET 1

Existing results



Inter-element spacing: $0.6 \lambda / 2$

Ambiguous spatial frequencies:

$$\arg z_1 = 0, \quad \arg z_2 = 1.0241,$$

$$\arg z_3 = 1.7312, \quad \arg z_4 = 2.6064,$$

$$\arg z_5 = -2.6064,$$

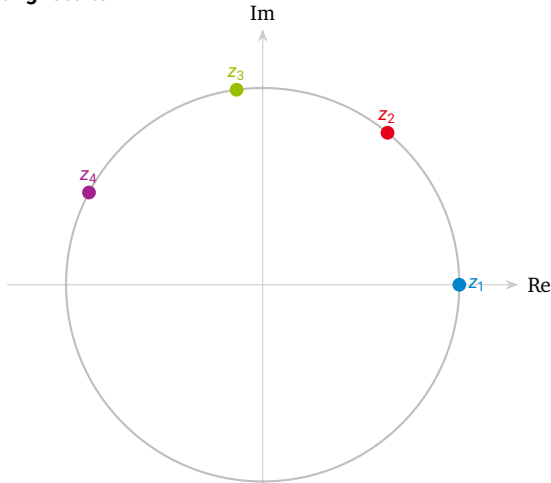
Any 4 columns of matrix

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & z_2^2 & z_3^2 & z_4^2 & z_5^2 \\ 1 & z_2^5 & z_3^5 & z_4^5 & z_5^5 \\ 1 & z_2^7 & z_3^7 & z_4^7 & z_5^7 \end{bmatrix}$$

are linearly dependent.

NON-UNIFORM AMBIGUITY SET 2

Existing results



Inter-element spacing: $0.6 \lambda / 2$

Ambiguous spatial frequencies:

$$\arg z_1 = 0, \quad \arg z_2 = 0.8830,$$

$$\arg z_3 = 1.7038, \quad \arg z_4 = 2.6536,$$

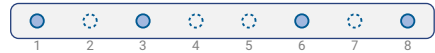
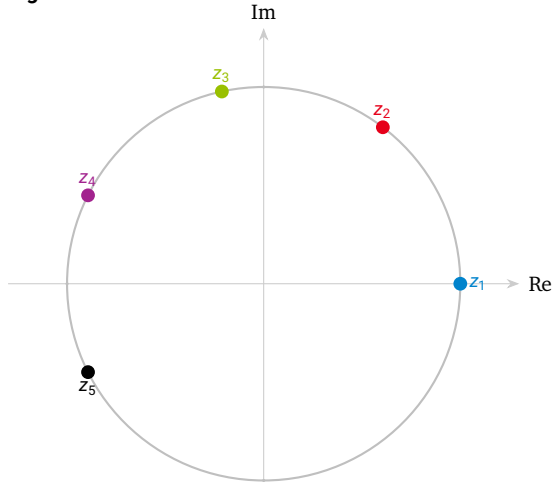
The columns of matrix

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & z_2^2 & z_3^2 & z_4^2 \\ 1 & z_2^5 & z_3^5 & z_4^5 \\ 1 & z_2^7 & z_3^7 & z_4^7 \end{bmatrix}$$

are linearly dependent.

NON-UNIFORM AMBIGUITY SET 3

Existing results



Inter-element spacing: $0.6 \lambda / 2$

Ambiguous spatial frequencies:

$$\arg z_1 = 0, \quad \arg z_2 = 0.9197,$$

$$\arg z_3 = 1.7851, \quad \arg z_4 = 2.6756,$$

$$\arg z_5 = -2.6756,$$

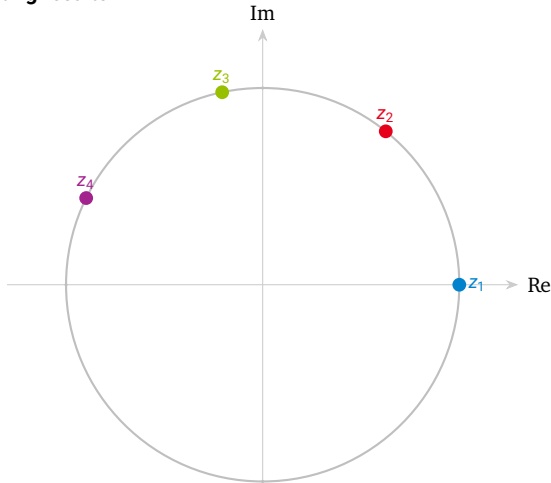
Any 4 columns of matrix

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & z_2^2 & z_3^2 & z_4^2 & z_5^2 \\ 1 & z_2^5 & z_3^5 & z_4^5 & z_5^5 \\ 1 & z_2^7 & z_3^7 & z_4^7 & z_5^7 \end{bmatrix}$$

are linearly dependent.

NON-UNIFORM AMBIGUITY SET 4

Existing results



Inter-element spacing: $0.6 \lambda / 2$

Ambiguous spatial frequencies:

$$\arg z_1 = 0, \quad \arg z_2 = 0.8944,$$

$$\arg z_3 = 1.7783, \quad \arg z_4 = 2.6853,$$

The columns of matrix

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & z_2^2 & z_3^2 & z_4^2 \\ 1 & z_2^5 & z_3^5 & z_4^5 \\ 1 & z_2^7 & z_3^7 & z_4^7 \end{bmatrix}$$

are linearly dependent.

NON-UNIFORM AMBIGUITIES

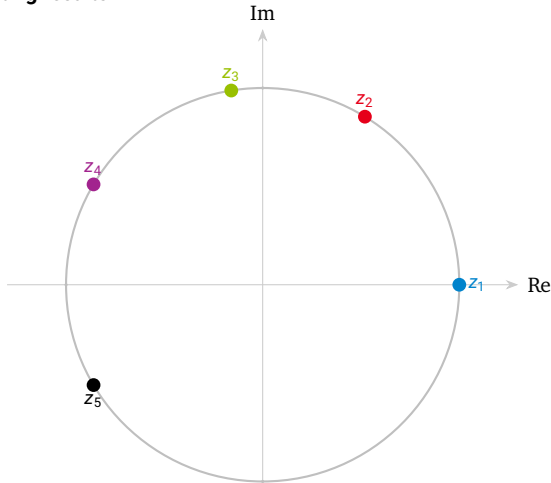
Existing results

Why does the reversed procedure work?

- The **number of distinct ambiguity sets** and **nullspace dimensions** are sufficiently **large** (as we will later see).
- The constructed polynomial has **centro-Hermitian coefficients**, which is a necessary condition for constructing polynomials with **all roots on the unit-circle**.
- The authors themselves have recognized that:
 - a sufficiently large number of roots on the unit-circle may not be obtained for the constructed polynomials.
 - the **ambiguity characterization is not exhaustive**, i.e. many other ambiguity generator sets exist.

NON-UNIFORM AMBIGUITY SET 1

Existing results



Inter-element spacing: $0.6 \lambda / 2$

Ambiguous spatial frequencies:

$$\arg z_1 = 0, \quad \arg z_2 = 1.0241,$$

$$\arg z_3 = 1.7312, \quad \arg z_4 = 2.6064,$$

$$\arg z_5 = -2.6064,$$

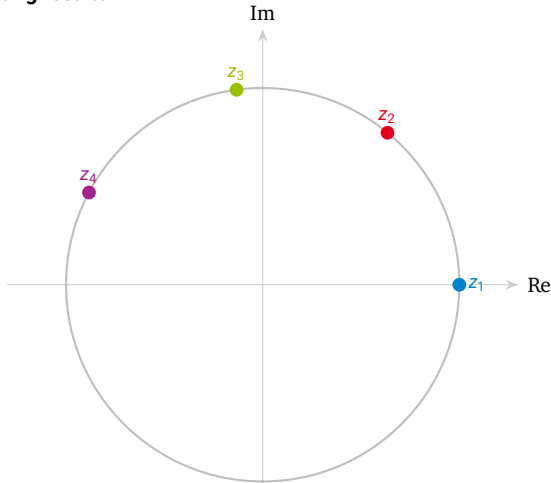
Any 4 columns of matrix

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & z_2^2 & z_3^2 & z_4^2 & z_5^2 \\ 1 & z_2^5 & z_3^5 & z_4^5 & z_5^5 \\ 1 & z_2^7 & z_3^7 & z_4^7 & z_5^7 \end{bmatrix}$$

are linearly dependent.

NON-UNIFORM AMBIGUITY SET 2

Existing results



Inter-element spacing: $0.6 \lambda / 2$

Ambiguous spatial frequencies:

$$\arg z_1 = 0, \quad \arg z_2 = 0.8830,$$

$$\arg z_3 = 1.7038, \quad \arg z_4 = 2.6536,$$

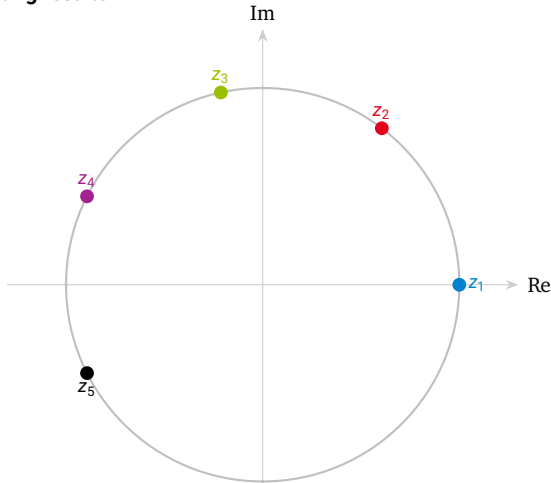
The columns of matrix

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & z_2^2 & z_3^2 & z_4^2 \\ 1 & z_2^5 & z_3^5 & z_4^5 \\ 1 & z_2^7 & z_3^7 & z_4^7 \end{bmatrix}$$

are linearly dependent.

NON-UNIFORM AMBIGUITY SET 3

Existing results



Inter-element spacing: $0.6 \lambda / 2$

Ambiguous spatial frequencies:

$$\arg z_1 = 0, \quad \arg z_2 = 0.9197,$$

$$\arg z_3 = 1.7851, \quad \arg z_4 = 2.6756,$$

$$\arg z_5 = -2.6756,$$

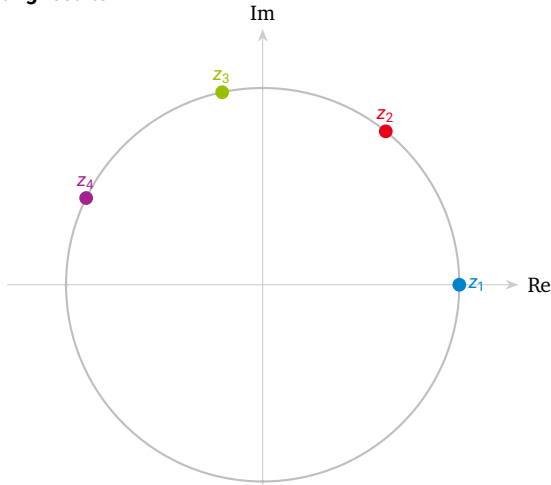
Any 4 columns of matrix

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & z_2^2 & z_3^2 & z_4^2 & z_5^2 \\ 1 & z_2^5 & z_3^5 & z_4^5 & z_5^5 \\ 1 & z_2^7 & z_3^7 & z_4^7 & z_5^7 \end{bmatrix}$$

are linearly dependent.

NON-UNIFORM AMBIGUITY SET 4

Existing results



Inter-element spacing: $0.6 \lambda / 2$

Ambiguous spatial frequencies:

$$\arg z_1 = 0, \quad \arg z_2 = 0.8944,$$

$$\arg z_3 = 1.7783, \quad \arg z_4 = 2.6853,$$

The columns of matrix

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & z_2^2 & z_3^2 & z_4^2 \\ 1 & z_2^5 & z_3^5 & z_4^5 \\ 1 & z_2^7 & z_3^7 & z_4^7 \end{bmatrix}$$

are linearly dependent.

SPARSE ARRAY ANALYSIS: SCHUR POLYNOMIAL APPROACH

Existing results

In Matter et al^[12], we take an alternative approach based on:

- The construction of a **Schur polynomial** based on the Young tableaux and the analytic factorization to determine the roots $z_1, \dots, z_L$.
- The formulation of a **mixed-integer program** under the roots of unity assumption (**discretized root locations on the unit-circle**).

Let $\mathbf{A}_{\mathcal{I}}(z_1, \dots, z_{M_{\mathcal{I}}})$ denote a $M_{\mathcal{I}} \times M_{\mathcal{I}}$ generalized Vandermonde matrix. Similarly, let $\mathbf{A}_{M_{\mathcal{I}}}(z_1, \dots, z_{M_{\mathcal{I}}})$ denote a $M_{\mathcal{I}} \times M_{\mathcal{I}}$ (classical) Vandermonde matrix.

Furthermore, let

$$V_{M_{\mathcal{I}}}(z_1, \dots, z_{M_{\mathcal{I}}}) = \det(\mathbf{A}_{M_{\mathcal{I}}}(z_1, \dots, z_{M_{\mathcal{I}}})) = \prod_{1 \leq i < k \leq M_{\mathcal{I}}} (z_k - z_i)$$

denote the classical Vandermonde determinant (which does not vanish as long as the generators $z_i \neq z_k$ for all pairs k, i).

[12] F. Matter et al., "Ambiguities in DoA Estimation With Linear Arrays", 2022

SPARSE ARRAY ANALYSIS: SCHUR POLYNOMIAL APPROACH

Existing results

Define

$$V_{\mathcal{I}}(z_1, \dots, z_{M_{\mathcal{I}}}) = \det(\mathbf{A}_{\mathcal{I}}(z_1, \dots, z_{M_{\mathcal{I}}}))$$

as the generalized Vandermonde determinant.

Then it is well known^[13] that $V_{M_{\mathcal{I}}}(z_1, \dots, z_{M_{\mathcal{I}}})$ divides $V_{\mathcal{I}}(z_1, \dots, z_{M_{\mathcal{I}}})$.

The ratio

$$S_{\mathcal{I}}(z_1, \dots, z_{M_{\mathcal{I}}}) = \frac{V_{\mathcal{I}}(z_1, \dots, z_{M_{\mathcal{I}}})}{V_{M_{\mathcal{I}}}(z_1, \dots, z_{M_{\mathcal{I}}})} = \sum_{R \in R_{\mathcal{I}}} \prod_{i=1}^{M_{\mathcal{I}}} z_i^{\alpha_i(R)}$$

is the so-called **Schur polynomial** that must vanish if $\mathbf{A}_{\mathcal{I}}(z_1, \dots, z_{M_{\mathcal{I}}})$ drops rank, i.e., $S_{\mathcal{I}}(z_1, \dots, z_{M_{\mathcal{I}}}) = 0$.

Here $R_{\mathcal{I}}$ is the set of all **Standard Young tableaux** with entries in $\{1, \dots, M\}$, and $\alpha_i(R)$ denotes integers that can be determined from the corresponding Young tableaux^[12].

[13] S. De Marchi, "Polynomials arising in factoring generalized Vandermonde determinants: an algorithm for computing their coefficients", 2001

[12] F. Matter et al., "Ambiguities in DoA Estimation With Linear Arrays", 2022

SPARSE ARRAY ANALYSIS: SCHUR POLYNOMIAL APPROACH

Existing results

For example for 3 sensor sparse (thinned) array obtained from a 5 sensor ULA the Schur polynomial becomes

$$S_{\mathcal{I}}(z_1, \dots, z_{M_{\mathcal{I}}}) = z_1 z_2 z_3 (z_1^2 + z_1 z_2 + z_2^2 + z_1 z_3 + z_2 z_3 + z_3^2).$$



$\mathbf{a}_{\text{ULA}}$

$$\begin{bmatrix} 1 & z^1 & z^2 & z^3 & z^4 \end{bmatrix}^T$$

thinned →



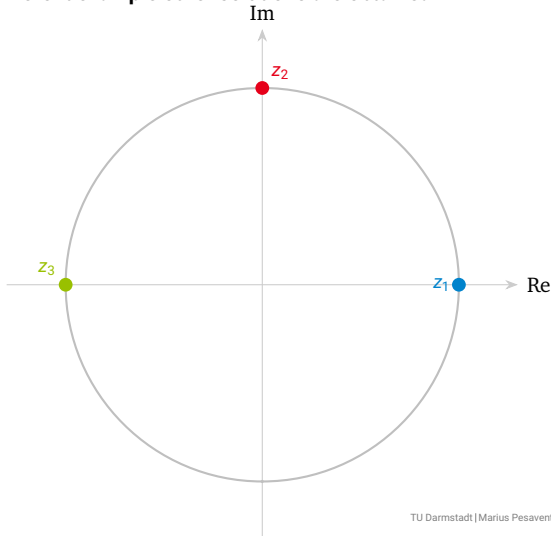
$\mathbf{a}_{\text{sparse}}$

$$\begin{bmatrix} 1 & z^1 & \square & \square & z^4 \end{bmatrix}^T$$



ALGEBRAIC SOLUTION YIELDS KNOWN SOLUTION FAMILY

but in a different example other solutions are obtained



Uniform generator set:

$$\arg z_1 = 0, \quad \arg z_2 = \frac{\pi}{2},$$

$$\arg z_3 = \pi,$$

Singular array steering matrix

$$\mathbf{A}(z_1, z_2, z_3) = \begin{bmatrix} 1 & 1 & 1 \\ 1 & e^{j\frac{\pi}{2}} & e^{j\pi} \\ 1 & e^{j4\frac{\pi}{2}} & e^{j4\pi} \end{bmatrix}$$

are linearly dependent.

MIXED INTEGER APPROACH

Existing results

- Determining algebraically all generators $z_1, \dots, z_L$ on the unit-circle, for which the Schur polynomial vanishes, may not always be possible.
- Alternatively, select solution set from the sets of roots of unity ($z^K = 1$), i.e., the sets

$$\{1, e^{j\frac{2\pi}{K}}, \dots, e^{j\frac{2\pi(K-1)}{K}}\}, \quad (K = 2, \dots, M)$$

using a mixed integer programming approach.



ROOTS OF UNITY

MIXED INTEGER APPROACH

- This **discretization** approach is also motivated by the observation that the **uniform ambiguities** belong to this class.
- This approach has revealed **new ambiguity sets** that have not been discovered before, however, at a **high computational cost**.
- The approach **does not scale well** with the aperture M .

LIMITATIONS OF EXISTING ANALYSIS APPROACHES

- Existing analysis approaches are **limited to array steering matrices with the same number of sources as sensors** ($L = M_{\mathcal{I}}$), i.e., **square generalized Vandermonde** matrices.
- In practice, however, we know that all thinned ULAs exhibit ambiguities and we want to **determine all ambiguity sets** for $L \leq M_{\mathcal{I}}$ or **prove** that for a **given source number L** the array is **free of ambiguities**.
- Existing analysis approaches **do not fully characterize all ambiguity sets**. In fact, it is known, that each approach is only characterizing particular subsets of ambiguities.
- Existing analysis approaches **do not scale** well with the **aperture M** and the **number of sensors $M_{\mathcal{I}}$** in the sparse array, or the omitted sensors $M_{\bar{\mathcal{I}}}$.
- Existing ambiguity analysis approaches are not constructive in the sense that they allow to draw conclusions on the **design of sparse arrays** that are free from ambiguities.

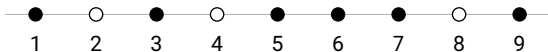
Sparse array analysis

Generalized Vandermonde rank conditions



SCALABLE AMBIGUITY CHARACTERIZATION APPROACH

for thinned uniform linear arrays



- We consider an arbitrary source number $L < M_I$, i.e., the general tall rather than the square steering matrix case:

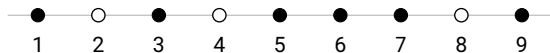
$$\mathbf{y}_I = \mathbf{A}_I(\mathbf{z}_1, \dots, \mathbf{z}_L) \mathbf{x} + \text{noise}$$

- $\mathbf{A}_I(\mathbf{z}_1, \dots, \mathbf{z}_L)$ is a generalized Vandermonde obtained from thinning a full Vandermonde:

$$\mathbf{A}_I(\mathbf{z}_1, \mathbf{z}_2, \mathbf{z}_3, \dots, \mathbf{z}_5) = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ \mathbf{z}_1 & \mathbf{z}_2 & \mathbf{z}_3 & \mathbf{z}_4 & \mathbf{z}_5 \\ \mathbf{z}_1^2 & \mathbf{z}_2^2 & \mathbf{z}_3^2 & \mathbf{z}_4^2 & \mathbf{z}_5^2 \\ \mathbf{z}_1^3 & \mathbf{z}_2^3 & \mathbf{z}_3^3 & \mathbf{z}_4^3 & \mathbf{z}_5^3 \\ \mathbf{z}_1^4 & \mathbf{z}_2^4 & \mathbf{z}_3^4 & \mathbf{z}_4^4 & \mathbf{z}_5^4 \\ \mathbf{z}_1^5 & \mathbf{z}_2^5 & \mathbf{z}_3^5 & \mathbf{z}_4^5 & \mathbf{z}_5^5 \\ \mathbf{z}_1^6 & \mathbf{z}_2^6 & \mathbf{z}_3^6 & \mathbf{z}_4^6 & \mathbf{z}_5^6 \\ \mathbf{z}_1^7 & \mathbf{z}_2^7 & \mathbf{z}_3^7 & \mathbf{z}_4^7 & \mathbf{z}_5^7 \\ \mathbf{z}_1^8 & \mathbf{z}_2^8 & \mathbf{z}_3^8 & \mathbf{z}_4^8 & \mathbf{z}_5^8 \end{bmatrix}.$$

- Generators $\mathbf{z}_\ell = e^{j2\pi\theta_\ell}$ are on the unit-circle and $\theta_\ell = \frac{2\pi d}{\lambda} \cos \vartheta$ are the electrical angles (direction cosines) corresponding to the impinging directions ϑ .

THINNED POLYNOMIAL EXAMPLE: $L = 5$



Fundamental questions:

1. What is the **maximum number of sources** L , for which $\mathbf{A}_{\mathcal{I}}(z_1, z_2, z_3 \dots, z_L)$ is always full rank? This corresponds to determining the Kruskal rank = $\text{spark}(\mathbf{A}_{\mathcal{I}}(z_1, z_2, z_3 \dots, z_L)) - 1$.
2. For fixed number L , **what are the generator sets** $\{z_1, \dots, z_L\}$ for which $\mathbf{A}_{\mathcal{I}}(z_1, z_2, z_3 \dots, z_L)$ becomes rank deficient? The corresponding scenarios are not resolvable!

To answer these questions define the Toeplitz matrix $\mathcal{T}(\mathbf{g})$ and the complementary submatrix $\mathcal{T}_{\bar{\mathcal{I}}}(\mathbf{g})$:

$$\mathcal{T}(\mathbf{g}) = \begin{bmatrix} g_0 & g_1 & g_2 & g_2^* & g_1^* & g_0^* & 0 & 0 & 0 \\ 0 & g_0 & g_1 & g_2 & g_2^* & g_1^* & g_0^* & 0 & 0 \\ 0 & 0 & g_0 & g_1 & g_2 & g_2^* & g_1^* & g_0^* & 0 \\ 0 & 0 & 0 & g_0 & g_1 & g_2 & g_2^* & g_1^* & g_0^* \end{bmatrix} \Rightarrow \mathcal{T}_{\bar{\mathcal{I}}}(\mathbf{g}) = \begin{bmatrix} g_1 & g_2^* & 0 \\ g_0 & g_2 & 0 \\ 0 & g_1 & g_0^* \\ 0 & g_0 & g_1^* \end{bmatrix}$$

MAIN RESULT

For the complete proof see pre-print: <http://arxiv.org/abs/2606.00360>

Theorem 1

For the thinned array given above with $M = 9$, $L = 5$ and the sensor set $\mathcal{I} = \{1, 3, 5, 6, 7, 9\}$, the coefficient vector $\mathbf{g} = [g_0, g_1, g_2, g_2^*, g_1^*, g_0^*]^\top$, with bounding coefficient $g_0 \neq 0$, for which the complementary submatrix

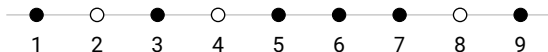
$$\mathcal{T}_{\bar{\mathcal{I}}}(\mathbf{g}) = \begin{bmatrix} g_1 & g_2^* & 0 \\ g_0 & g_2 & 0 \\ 0 & g_1 & g_0^* \\ 0 & g_0 & g_1^* \end{bmatrix}$$

becomes rank deficient, yields a full polynomial

$$P_{\mathbf{g}}(t) = g_0 + g_1 z + g_2 z^2 + g_2^* z^3 + g_1^* z^4 + g_0^* z^5 = k \prod_{\ell=1}^5 (z - z_{\ell}).$$

with roots $z_1, \dots, z_5$ and constant k , for which $\mathbf{A}_{\mathcal{I}}(z_1, z_2, z_3, \dots, z_5)$ becomes rank deficient (if all roots are distinct and on the unit-circle).

SPARSE ARRAY EXAMPLE



- Setting $g_0 = 1$ and performing **Gaussian elimination with pivoting** yields

$$\mathcal{T}_{\bar{\mathcal{I}}}(\mathbf{g}) = \begin{bmatrix} g_1 & g_2^* & 0 \\ g_0 & g_2 & 0 \\ 0 & g_1 & g_0^* \\ 0 & g_0 & g_1^* \end{bmatrix} \stackrel{\text{row}}{\equiv} \begin{bmatrix} 1 & g_2 & 0 \\ 0 & 1 & g_1^* \\ 0 & 0 & g_1^* |g_2| (e^{j\phi_2} g_1 - e^{-j\phi_2}) \\ 0 & 0 & 1 - |g_1|^2 \end{bmatrix}.$$

which is **rank deficient** for:

- $g_2 = 0$ and $|g_1| = 1$:

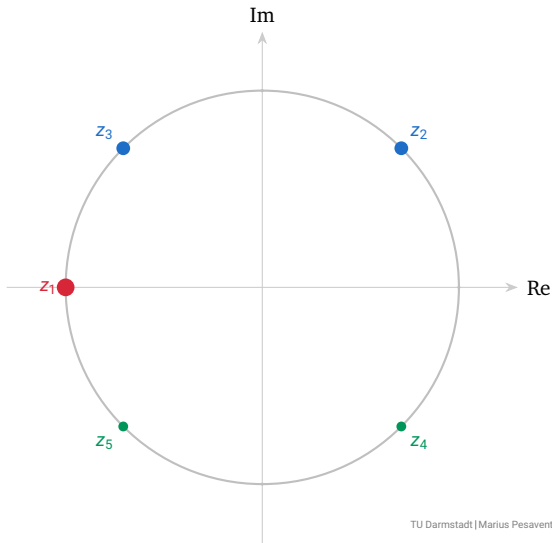
$$P_g(z) = 1 + z + z^4 + z^5 = (z + 1) \prod_{p=1}^4 (z - e^{j(2p-1)\pi/4}).$$

- $g_2 = |g_2| e^{j\phi_2} \neq 0$ and $g_1 = e^{-j2\phi_2}$:

$$P_g(z) = (ze^{-j2\phi_2} + 1)(e^{j\phi_2} z^2 + \alpha)(e^{j\phi_2} z^2 + \alpha^*)$$

- By Theorem 1 the **array steering matrix** $\mathbf{A}_{\mathcal{I}}(z_1, \dots, z_5) \in \mathbb{C}^{7 \times 5}$ is **rank deficient**.

SOLUTION 1: AMBIGUOUS GENERATOR SET



Fixed root locations

$$\arg z_1 = -\pi,$$

$$\arg z_{2,3} = \frac{\pi}{4}, \frac{3\pi}{4},$$

$$\arg z_{4,5} = \frac{7\pi}{4}, \frac{5\pi}{4}.$$

All five lie on the unit-circle.

SOLUTION 2: AMBIGUOUS GENERATOR SETS

The five roots

$$z_1 = -e^{j2\phi_2}, \quad z_{2,3} = \pm j e^{j(\psi - \phi_2)/2},$$

$$z_{4,5} = \pm j e^{-j(\psi + \phi_2)/2}$$

all lie on the unit-circle, with

$$\phi_2 \in [-\pi, \pi], \quad \psi = \arccos(|g_2|/2),$$

and $|g_2| \in [0, 2]$.



USE A CONVENIENT TOEPLITZ MATRIX REPRESENTATION

Sketch of proof of Theorem 1. For details see: <http://arxiv.org/abs/2606.00360>

For full Vandermonde matrix $\mathbf{A}$

$$\begin{array}{c}
 \begin{array}{c} M \\ \text{---} \\ \boxed{\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \dots \\ \text{---} \end{array}} \\ \text{---} \\ M-L \end{array} \\
 \mathcal{T}(\mathbf{g})
 \end{array}
 \times
 \begin{array}{c}
 L \\ \boxed{} \\ M \\ \mathbf{A}
 \end{array}
 =
 \begin{array}{c}
 L \\ \boxed{0} \\ L
 \end{array}$$

- Rows of $\mathcal{T}(\mathbf{g})$ built from shifted parameter vector $\mathbf{g}$
- These rows form a basis of the orthogonal complement of $\mathcal{R}(\mathbf{A})$, i.e $\mathcal{R}(\mathcal{T}^T(\mathbf{g})) = \mathcal{R}^\perp(\mathbf{A})$



TOEPLITZ STRUCTURE OF $\mathcal{T}(g)$

Sketch of proof of Theorem 1. For details see: <http://arxiv.org/abs/2606.00360>

even: $L=2K$

$$\mathcal{T}(g) = \begin{bmatrix} \overset{1}{g_0} & \overset{2}{g_1} & \overset{3}{g_2} & \overset{4}{g_3} & \cdots & \overset{K-2}{g_{K-3}} & \overset{K-1}{g_{K-2}} & \overset{K}{g_{K-1}} & \overset{K+1}{g_K} & \overset{K+2}{g_{K-1}^*} & \overset{K+3}{g_{K-2}^*} & \overset{K+4}{g_{K-3}^*} & \cdots & \overset{L-2}{g_3^*} & \overset{L-1}{g_2^*} & \overset{L}{g_1^*} & \overset{L+1}{g_0^*} & \overset{L+2}{0} & \cdots & \overset{M}{0} \\ 0 & g_0 & g_1 & g_2 & g_3 & \cdots & g_{K-3} & g_{K-2} & g_{K-1} & g_K & g_{K-1}^* & g_{K-2}^* & g_{K-3}^* & \cdots & g_3^* & g_2^* & g_1^* & g_0^* & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & g_0 & g_1 & g_2 & g_3 & \cdots & g_{K-3} & g_{K-2} & g_{K-1} & g_K & g_{K-1}^* & g_{K-2}^* & g_{K-3}^* & \cdots & g_3^* & g_2^* & g_1^* & g_0^* \\ \underset{1}{\cdots} & \cdots & \underset{M-L-1}{\cdots} & \underset{M-L}{\cdots} & \underset{M-L+1}{\cdots} & \underset{M-L+2}{\cdots} & \underset{M-L+3}{\cdots} & \cdots & \underset{M-K-2}{\cdots} & \underset{M-K-1}{\cdots} & \underset{M-K}{\cdots} & \underset{M-K+1}{\cdots} & \underset{M-K+2}{\cdots} & \underset{M-K+3}{\cdots} & \underset{M-K+4}{\cdots} & \cdots & \underset{M-3}{\cdots} & \underset{M-2}{\cdots} & \underset{M-1}{\cdots} & \underset{M}{\cdots} \end{bmatrix}$$

odd: $L=2K+1$

$$\mathcal{T}(g) = \begin{bmatrix} \overset{1}{g_0} & \overset{2}{g_1} & \overset{3}{g_2} & \overset{4}{g_3} & \cdots & \overset{K-2}{g_{K-3}} & \overset{K-1}{g_{K-2}} & \overset{K}{g_{K-1}} & \overset{K+1}{g_K} & \overset{K+2}{g_{K-1}^*} & \overset{K+3}{g_{K-2}^*} & \overset{K+4}{g_{K-3}^*} & \cdots & \overset{L-2}{g_3^*} & \overset{L-1}{g_2^*} & \overset{L}{g_1^*} & \overset{L+1}{g_0^*} & \overset{L+2}{0} & \cdots & \overset{M}{0} \\ 0 & g_0 & g_1 & g_2 & g_3 & \cdots & g_{K-3} & g_{K-2} & g_{K-1} & g_K & g_{K-1}^* & g_{K-2}^* & g_{K-3}^* & \cdots & g_3^* & g_2^* & g_1^* & g_0^* & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & g_0 & g_1 & g_2 & g_3 & \cdots & g_{K-3} & g_{K-2} & g_{K-1} & g_K & g_{K-1}^* & g_{K-2}^* & g_{K-3}^* & \cdots & g_3^* & g_2^* & g_1^* & g_0^* \\ \underset{1}{\cdots} & \cdots & \underset{M-L-1}{\cdots} & \underset{M-L}{\cdots} & \underset{M-L+1}{\cdots} & \underset{M-L+2}{\cdots} & \underset{M-L+3}{\cdots} & \cdots & \underset{M-K-3}{\cdots} & \underset{M-K-2}{\cdots} & \underset{M-K-1}{\cdots} & \underset{M-K}{\cdots} & \underset{M-K+1}{\cdots} & \underset{M-K+2}{\cdots} & \underset{M-K+3}{\cdots} & \cdots & \underset{M-3}{\cdots} & \underset{M-2}{\cdots} & \underset{M-1}{\cdots} & \underset{M}{\cdots} \end{bmatrix}$$



THINNED VANDERMONDE $\mathbf{A}_{\mathcal{I}}$ WITH A RANK DEFICIENCY OF D

Sketch of proof of Theorem 1. For details see: <http://arxiv.org/abs/2606.00360>

$$\begin{array}{ccccc}
 & M_{\mathcal{I}} & & L & \\
 & & & & L \\
 M_{\mathcal{I}} - L + D & \begin{array}{|c|} \hline \text{Purple Box} \\ \hline \end{array} & \times & \begin{array}{|c|} \hline \text{Yellow Box} \\ \hline \end{array} & = & \begin{array}{|c|} \hline \text{White Box} \\ \hline \end{array} \\
 & \mathbf{K}_{\mathcal{I}}^T & & \mathbf{A}_{\mathcal{I}} & & \mathbf{0}
 \end{array}$$

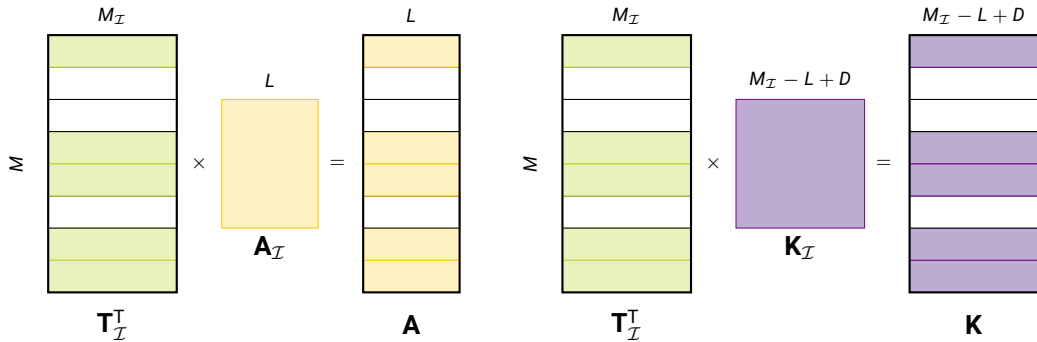
$\mathbf{A}_{\mathcal{I}}$ has a rank deficiency of D



There exists a full column rank matrix $\mathbf{K}_{\mathcal{I}}$ in orthogonal range space $\mathcal{R}^{\perp}(\mathbf{A}_{\mathcal{I}})$.

INSERTING ZERO ROWS VIA SELECTION MATRIX $\mathbf{T}_{\mathcal{I}}^T$

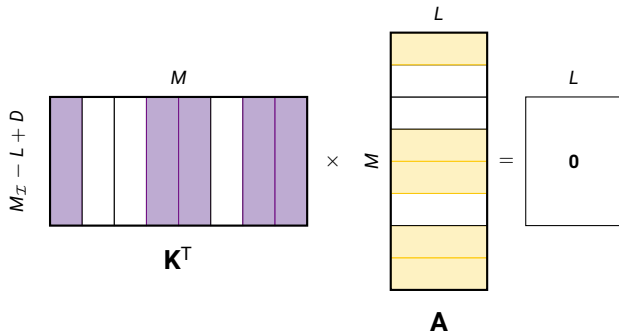
Sketch of proof of Theorem 1. For details see: <http://arxiv.org/abs/2606.00360>



The same selector $\mathcal{T}_{\mathcal{I}}^T$ inserts zero rows at $\bar{\mathcal{I}}$ into both $\mathbf{A}_{\mathcal{I}} \rightarrow \mathbf{A}$ and $\mathbf{K}_{\mathcal{I}} \rightarrow \mathbf{K}$
 This allows for defining a matrix $\mathbf{K} = \mathbf{T}_{\mathcal{I}}^T \mathbf{K}_{\mathcal{I}}$ such that $\mathbf{K}^T \mathbf{A} = \mathbf{0}$!

INDUCED ROW SPARSITY PRESERVES NULLSPACE

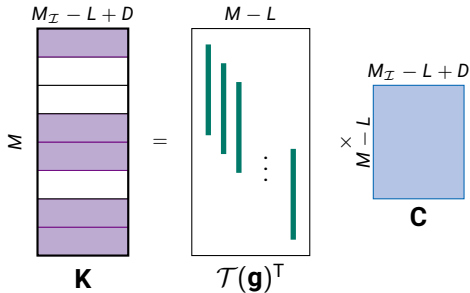
Sketch of proof of Theorem 1. For details see: <http://arxiv.org/abs/2606.00360>



$\mathbf{K}^T \mathbf{A} = \mathbf{0}$, with zero columns of $\mathbf{K}^T$ and zero rows of $\mathbf{A}$ at the same positions. $\Rightarrow \mathcal{R}(\mathbf{K}) \subset \mathcal{R}^\perp(\mathbf{A})$

K LIVES IN THE ROW SPACE OF $\mathcal{T}(\mathbf{g})$

Sketch of proof of Theorem 1. For details see: <http://arxiv.org/abs/2606.00360>



$$\mathbf{K} = \mathcal{T}(\mathbf{g})^T \mathbf{C}$$

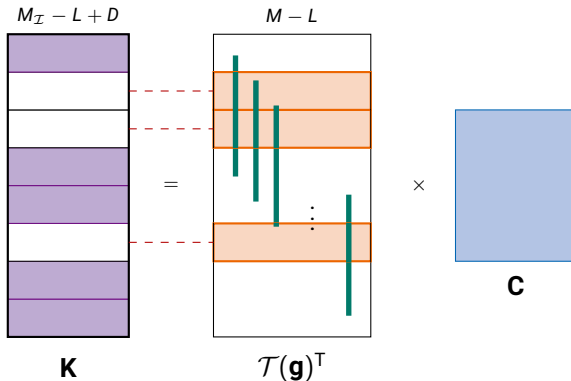
$$\mathcal{R}(\mathbf{K}) \subset \mathcal{R}(\mathcal{T}(\mathbf{g})^T) = \mathcal{R}^\perp(\mathbf{A})$$



There exists a full column rank $\mathbf{C}$ such that $\mathbf{K} = \mathcal{T}(\mathbf{g})^T \mathbf{C}$.

MARKING THE ROWS $\bar{\mathcal{I}}$ ON BOTH SIDES

Sketch of proof of Theorem 1. For details see: <http://arxiv.org/abs/2606.00360>



The rows belonging to $\bar{\mathcal{I}}$ (dashed) occupy the same *positions* in $\mathbf{K}$ and in $\mathcal{T}(\mathbf{g})^T$.

EXTRACTING THOSE ROWS YIELDS THE KEY IDENTITY

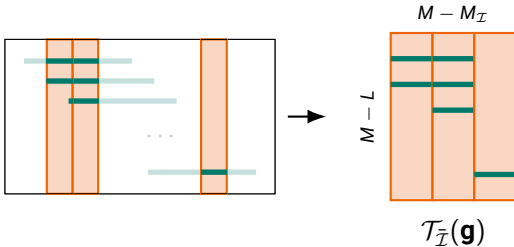
Sketch of proof of Theorem 1. For details see: <http://arxiv.org/abs/2606.00360>

$$\begin{array}{c}
 \left. \mathbf{K}_{\bar{\mathcal{I}}} = \mathbf{0} \right\} \begin{array}{c} M_{\mathcal{I}} - L + D \\ \text{[Matrix with alternating purple and white rows]} \\ \mathbf{K} \end{array} = \begin{array}{c} M - L \\ \text{[Matrix with orange highlighted rows and greyed-out columns]} \\ \mathcal{T}(\mathbf{g})^T \end{array} \times \begin{array}{c} \text{[Blue square matrix]} \\ \mathbf{C} \end{array}
 \end{array}$$

Only the marked rows matter here; the rest of $\mathbf{K}$ and $\mathcal{T}(\mathbf{g})^T$ (greyed out) is irrelevant

FUNDAMENTAL RANK-DEFICIENCY CONDITION

Sketch of proof of Theorem 1. For details see: <http://arxiv.org/abs/2606.00360>



$$\begin{aligned} \mathcal{T}_{\bar{I}}(\mathbf{g})^T \mathbf{C} &= \mathbf{0} \\ \text{with } \mathbf{C} \text{ full column rank} \\ \Rightarrow \mathcal{T}_{\bar{I}}(\mathbf{g}) &\text{ must be rank deficient} \end{aligned}$$



$$\mathbf{A}_{\mathcal{I}}(z_1, \dots, z_L) \text{ is rank deficient}$$

MAIN RESULT

For the complete proof see pre-print: <http://arxiv.org/abs/2606.00360>

Theorem 1 (Rank Condition for Generalized Vandermonde Matrices)

Let $\mathbf{A}_{\mathcal{I}}(z_1, \dots, z_L) \in \mathbb{C}^{M_{\mathcal{I}} \times L}$ be a generalized Vandermonde matrix with index set $\mathcal{I}$ and $L \leq M_{\mathcal{I}} = |\mathcal{I}|$, and let $\mathbf{g} = [g_0, \dots, g_L]^T$ be the coefficient vector of the centro-Hermitian polynomial $P_{\mathbf{g}}(z) = \sum_{\ell=0}^L g_{\ell} z^{\ell}$ whose roots are $z_1, \dots, z_L$. Then $\mathbf{A}_{\mathcal{I}}(z_1, \dots, z_L)$ is rank deficient if and only if the complementary submatrix

$$\mathcal{T}_{\bar{\mathcal{I}}}(\mathbf{g}) = \mathcal{T}(\mathbf{g}) \mathbf{T}_{\bar{\mathcal{I}}}^T \in \mathbb{C}^{(M-L) \times (M-M_{\mathcal{I}})}$$

is rank deficient, i.e.,

$$\text{rank}(\mathcal{T}_{\bar{\mathcal{I}}}(\mathbf{g})) = M - M_{\mathcal{I}} - D$$

for some integer D with $1 \leq D < L$.

RANK INVARIANCE WITH RESPECT TO ROOT ROTATION

- The array polynomial is built directly from its L roots, all lying on the unit-circle:

$$P(z) = \prod_{\ell=1}^L (z - z'_\ell) = \sum_{\ell=0}^L g_\ell z^\ell, \quad g_L = 1$$

- Each g_ℓ is an elementary symmetric function $\varepsilon_k(z'_1, \dots, z'_L) = \sum_{1 \leq i_1 < \dots < i_k \leq L} z'_{i_1} \cdots z'_{i_k}$ of the roots

$$g_\ell = (-1)^{L-\ell} \varepsilon_{L-\ell}(z'_1, \dots, z'_L)$$

- Rotation of every root by the same shift ϕ : $z'_\ell \longrightarrow z'_\ell e^{j\phi}$
- Since ε_k contains sums of exactly k root factors, e.g., $z'_{\ell_1} z'_{\ell_2}, \dots, z'_{\ell_k}$, each coefficient inherits a phase:

$$g_\ell(\phi) = g_\ell(\phi = 0) e^{j(L-\ell)\phi}$$

- A common rotation of the roots becomes a **linear phase increment across the coefficient index**: the phase increment is absorbed in the polynomial coefficients.

ROTATING ROOTS VS PLAIN COEFFICIENTS

RESTORING CONJUGATE SYMMETRY UNDER ROTATION

- The polynomials from the roots are self-inversive (*Geometry of Polynomials*, 1966)^[19]

1. **Definition:** $g(z) = \sum_{\ell} b_{\ell} z^{\ell}$ is **self-inversive** if

$$b_{L-\ell} = u b_{\ell}^*, \quad |u| = 1.$$

2. **Roots on unit-circle** ($|z| = 1$) $\Rightarrow$ **self-inversive:** each root is its own mirror image under $z \mapsto 1/z^*$.
3. **Self-inversive** $\Rightarrow$ **conjugate-symmetric coefficients, up to u :**

Write $u = e^{-2j\alpha}$; rotating by $e^{j\alpha}$ gives $h_{\ell} = b_{\ell} e^{j\alpha}$ with $h_{L-\ell} = h_{\ell}^*$.

- **Rotating the roots** ($z_{\ell} \rightarrow z_{\ell} e^{j\phi}$) keeps $|z_{\ell}| = 1$, so u is unchanged. Counter-rotating by $e^{-jL\phi/2}$:

$$g_{\ell}(\phi) = g_{\ell}(0) e^{j\alpha} e^{j(\frac{L}{2}-\ell)\phi} \implies g_{L-\ell}(\phi) = g_{\ell}^*(\phi) \quad \forall \phi.$$

[19] M. Marden, "Geometry of Polynomials", 1966



ROTATING ROOTS VS COEFFICIENTS PRESERVING SYMMETRY



ROTATION INVARIANCE IN COMPLEMENTARY SUBMATRIX

Corollary 1 in pre-print <http://arxiv.org/abs/2606.00360>

It is easy to show that

Corollary 1

The *rotation invariance property* is *inherited* by the *complementary submatrix*

$$\text{rank}(\mathcal{T}_{\bar{\mathcal{I}}}(\mathbf{g})) = \text{rank}(\mathcal{T}_{\bar{\mathcal{I}}}(\mathbf{g}(\phi))).$$

CONJUGATE INVARIANCE OF AMBIGUOUS GENERATOR SETS

Corollary 2 in pre-print <http://arxiv.org/abs/2606.00360>

Corollary 2

The complementary submatrix $\mathcal{T}_{\bar{\mathcal{I}}}(\mathbf{g})$ is rank deficient if and only if $\mathcal{T}_{\bar{\mathcal{I}}}(\mathbf{g}^)$ is rank deficient.*

Hence, taking **complex conjugates** of all the coefficients g_ℓ does **not change** whether the associated Toeplitz submatrix loses rank.

This implies that if $\{z_1, \dots, z_L\}$ are the roots of $P_{\mathbf{g}}(z)$, then $\{z_1^*, \dots, z_L^*\}$ are the roots of $P_{\mathbf{g}^*}(z)$, since

$$P_{\mathbf{g}^*}(z) = (P_{\mathbf{g}}(z^*))^*.$$

CONJUGATING THE COMPLEMENTARY SUBMATRIX

Sketch of proof of Corollary 2: see pre-print <http://arxiv.org/abs/2606.00360>

Inspection of the definitions of $\mathcal{T}(\mathbf{g})$ (even and odd cases) shows that replacing each coefficient g_ℓ by g_ℓ^* replaces every entry of $\mathcal{T}(\mathbf{g})$ by its complex conjugate:

$$\mathcal{T}(\mathbf{g}^*) = \mathcal{T}(\mathbf{g})^*.$$

Since $\mathbf{T}_{\bar{\mathcal{I}}}$ is a *real* selection matrix, conjugation passes through it:

$$\mathcal{T}_{\bar{\mathcal{I}}}(\mathbf{g}^*) = \mathcal{T}(\mathbf{g}^*) \mathbf{T}_{\bar{\mathcal{I}}}^T = (\mathcal{T}(\mathbf{g}) \mathbf{T}_{\bar{\mathcal{I}}}^T)^* = \mathcal{T}_{\bar{\mathcal{I}}}(\mathbf{g})^*.$$

CONCLUSION

Sketch of proof of Corollary 2: see pre-print <http://arxiv.org/abs/2606.00360>

Since the **rank** of a matrix is **invariant under complex conjugation**,

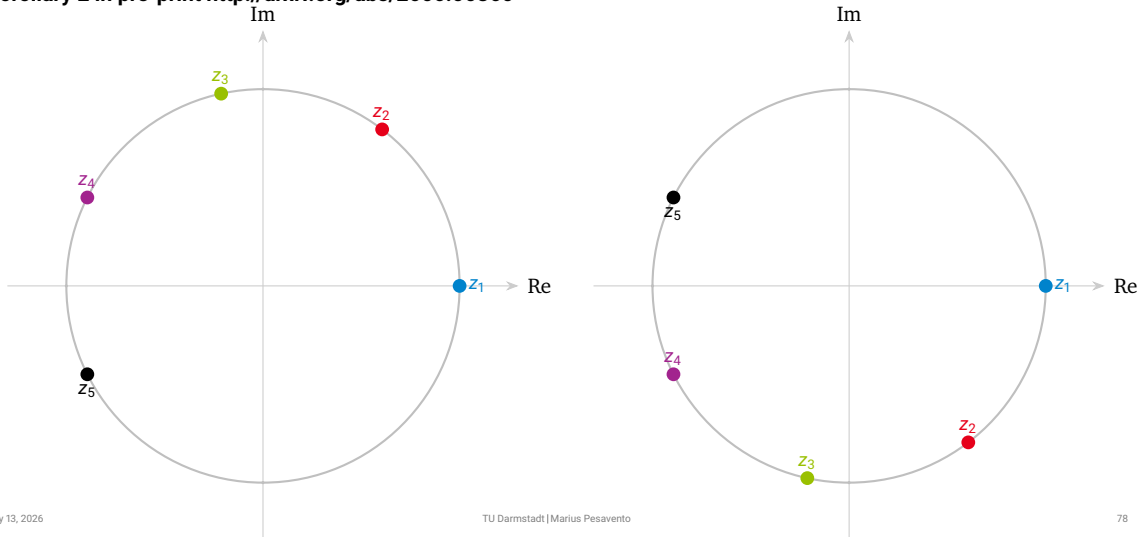
$$\text{rank}(\mathcal{T}_{\bar{\mathcal{I}}}(\mathbf{g}^*)) = \text{rank}(\mathcal{T}_{\bar{\mathcal{I}}}(\mathbf{g})^*) = \text{rank}(\mathcal{T}_{\bar{\mathcal{I}}}(\mathbf{g})).$$

Hence $\mathcal{T}_{\bar{\mathcal{I}}}(\mathbf{g})$ is **rank deficient** if and only if $\mathcal{T}_{\bar{\mathcal{I}}}(\mathbf{g}^*)$ is.



CONJUGATE INVARIANCE OF AMBIGUOUS GENERATOR SETS

Corollary 2 in pre-print <http://arxiv.org/abs/2606.00360>



VIOLATION OF SPATIAL SAMPLING THEOREM

Corollary 3 in pre-print <http://arxiv.org/abs/2606.00360>

Corollary 3

For a thinned ULA with $M_{\mathcal{I}} < M$ sensors and a number of sources $L \geq M_{\mathcal{I}}$, the thinned array steering matrix $\mathbf{A}_{\mathcal{I}}(z_1, \dots, z_L)$ is always ambiguous: there exists a set of distinct generators $\{z_1, \dots, z_L\}$ on the unit-circle for which $\mathbf{A}_{\mathcal{I}}(z_1, \dots, z_L)$ becomes rank deficient.

Sketch of proof

- It suffices to consider $L = M_{\mathcal{I}}$, for which the complementary submatrix $\mathcal{T}_{\bar{\mathcal{I}}}(\mathbf{g}) \in \mathbb{C}^{(M-L) \times (M-M_{\mathcal{I}})}$ is $M_{\bar{\mathcal{I}}} \times M_{\bar{\mathcal{I}}}$ square, with $M_{\bar{\mathcal{I}}} = M - M_{\mathcal{I}}$.
- Consider for $\mathbf{b} = \mathbf{e}_1 + \mathbf{e}_M \in \mathbb{B}^M$ the centro-Hermitian polynomial and its factorization

$$P_{\mathbf{b}}(z) = 1 + z^{M-1} = \prod_{m=1}^{M-1} (z - e^{j\pi(2m-1)/(M-1)}) = \underbrace{\left(\sum_{\ell=0}^{M_{\bar{\mathcal{I}}}-1} h_{\ell} z^{\ell} \right)}_{P_h(z)} \underbrace{\left(\sum_{\ell=0}^L g_{\ell} z^{\ell} \right)}_{P_g(z)}$$

- Hence $P_{\mathbf{b}}(z)$ has $M - 1$ roots on the unit-circle.

VIOLATION OF SPATIAL SAMPLING THEOREM

Corollary 3 in pre-print <http://arxiv.org/abs/2606.00360>

- The previous polynomial equation can be written in matrix form as

$$\mathbf{b}^\top = \underbrace{[1, 0, \dots, 0, 1]}_M = \underbrace{[h_0, h_1, \dots, h_1^*, h_0^*]}_{M_{\bar{\mathcal{I}}}} \mathcal{T}_{\bar{\mathcal{I}}}$$

- Right-multiplication by the complementary selection matrix $\mathbf{T}_{\bar{\mathcal{I}}}^\top$ and considering that by definition 1, $M \notin \bar{\mathcal{I}}$ such that $\mathbf{b}^\top \mathbf{T}_{\bar{\mathcal{I}}}^\top = \mathbf{0}^\top$ yields

$$\mathbf{b}^\top \mathbf{T}_{\bar{\mathcal{I}}}^\top = \mathbf{h}^\top \mathcal{T}(\mathbf{g}) \mathbf{T}_{\bar{\mathcal{I}}}^\top = \mathbf{h}^\top \mathcal{T}_{\bar{\mathcal{I}}}(\mathbf{g})$$

$$\Downarrow$$

$$\mathbf{0}^\top = \mathbf{h}^\top \mathcal{T}_{\bar{\mathcal{I}}}(\mathbf{g})$$

$$\Uparrow$$

$$\mathcal{T}_{\bar{\mathcal{I}}}(\mathbf{g}) \text{ rank deficient}$$

- The last equivalence follows from the fact that $\mathcal{T}_{\bar{\mathcal{I}}}(\mathbf{g})$ is square and $\mathbf{h} \neq \mathbf{0}$.

VIOLATION OF SPATIAL SAMPLING THEOREM

Corollary 3 in pre-print <http://arxiv.org/abs/2606.00360>

$$\begin{array}{c}
 \begin{array}{c} M \\ \hline \text{Diagram: A horizontal bar with green segments at both ends and orange segments in the middle.} \end{array} \\
 \mathbf{b}^T = [1, 0, \dots, 0, 1]^T
 \end{array}
 =
 \begin{array}{c}
 M - M_{\mathcal{I}} \\ \hline \text{Diagram: A horizontal bar with a red segment in the middle.} \\
 \mathbf{h}^T
 \end{array}
 \times
 \begin{array}{c}
 M \\ \hline \text{Diagram: A grid with orange columns and purple rows.} \\
 \mathcal{T}(\mathbf{g})
 \end{array}$$

SPATIAL SAMPLING THEOREM

- Corollary 3 states that, in a **thinned ULA**, the **number of columns L** of a thinned steering matrix $\mathbf{A}_{\mathcal{I}}(z_1, \dots, z_L)$ must be **strictly less than the number of physical sensors $M_{\mathcal{I}}$** , i.e. $L < M_{\mathcal{I}}$.
- This **contrasts the full ULA** case, where the Vandermonde determinant theorem guarantees full column rank of $\mathbf{A}_M(z_1, \dots, z_L)$ for any $L \leq M$ and any set of distinct generators.
- This result has previously been reported in (Manikas et al,1998)^[7] and (Abramovich,1999)^[8].

[7] A. Manikas and C. Proukakis, "Modeling and estimation of ambiguities in linear arrays", Aug. 1998

[8] Y. Abramovich et al., "Identifiability and manifold ambiguity in DOA estimation for nonuniform linear antenna arrays", Mar. 1999

SINGLE MISSING SENSOR

Example 1 of pre-print <http://arxiv.org/abs/2606.00360>



- For $M = 9$ and $L = M_{\mathcal{I}} = M - 1 = 8$ sources, the Toeplitz matrix is the single row

$$\mathcal{T}(\mathbf{g}) = [g_0, g_1, g_2, g_3, g_4, g_3^*, g_2^*, g_1^*, g_0^*],$$

where $g_0, \dots, g_3 \in \mathbb{C}$ with $g_0 \neq 0$ and $g_4 \in \mathbb{R}$.

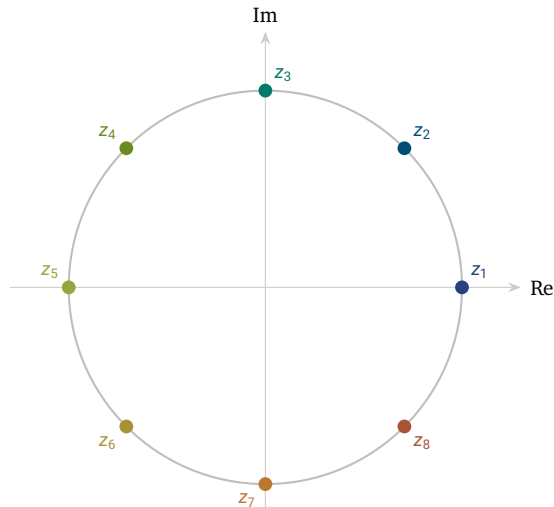
- The **complementary submatrix** reduces to the **scalar** $\mathcal{T}_{\bar{\mathcal{I}}}(\mathbf{g}) = g_{i-1}$, which vanishes for $g_{i-1} = 0$.
- It can be show that a corresponding polynomial

$$P_{\mathbf{g}}(z) = \sum_{\ell=0}^8 g_{\ell} z^{\ell}, \quad g_{i-1} = g_{8-i}^* = 0, \quad i \neq 1, 8$$

with all roots on the unit-circle always be found.

- By Theorem 1, **ambiguities always exist**.
- This is a special case of **Corollary 1**.

ALGEBRAIC SOLUTION BELONGS TO UNIFORM AMBIGUITY FAMILY



$$\mathbf{A}_{\mathcal{I}}(z_1, \dots, z_8) =$$

1	1	1	1	1	1	1	1
z_1	z_2	z_3	z_4	z_5	z_6	z_7	z_8
z_1^2	z_2^2	z_3^2	z_4^2	z_5^2	z_6^2	z_7^2	z_8^2
z_1^3	z_2^3	z_3^3	z_4^3	z_5^3	z_6^3	z_7^3	z_8^3
z_1^4	z_2^4	z_3^4	z_4^4	z_5^4	z_6^4	z_7^4	z_8^4
z_1^5	z_2^5	z_3^5	z_4^5	z_5^5	z_6^5	z_7^5	z_8^5
z_1^6	z_2^6	z_3^6	z_4^6	z_5^6	z_6^6	z_7^6	z_8^6
z_1^7	z_2^7	z_3^7	z_4^7	z_5^7	z_6^7	z_7^7	z_8^7
z_1^8	z_2^8	z_3^8	z_4^8	z_5^8	z_6^8	z_7^8	z_8^8

MAXIMUM-CONSECUTIVE-LAG ARRAYS

Corollary 4 in pre-print <http://arxiv.org/abs/2606.00360>

Corollary 4

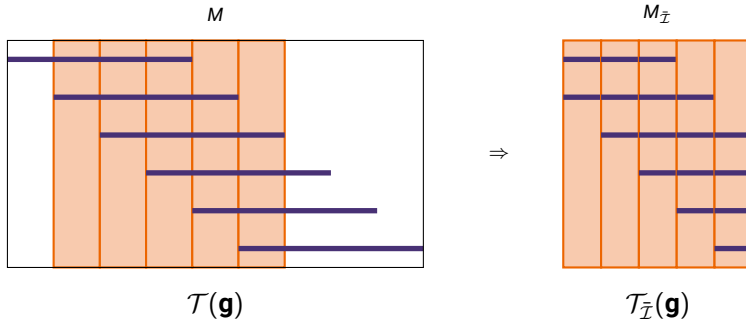
If the number of sensors in the thinned array exceeds the number of sources, i.e., $M_{\mathcal{I}} > L$, then there always exists a sparse sensor configuration, obtained by removing up to $M - M_{\mathcal{I}}$ sensors from the underlying M -element ULA, whose array manifold is unambiguous, i.e., $\mathbf{A}_{\mathcal{I}}(\mathbf{z}_1, \dots, \mathbf{z}_L) \in \mathbb{C}^{M_{\mathcal{I}} \times L}$ has full column rank for every set of distinct generators $\{\mathbf{z}_1, \dots, \mathbf{z}_L\}$.

Sketch of proof

- Consider the boundary case $M_{\mathcal{I}} = L + 1$ and choose $\bar{\mathcal{I}} = \{2, 3, \dots, M - L\}$, thus $\mathcal{I} = \{1, M - L + 1, M - L + 2, \dots, M\}$.
- The corresponding complementary submatrix $\mathcal{T}_{\bar{\mathcal{I}}}(\mathbf{g}) \in \mathbb{C}^{(M-L) \times (M-L-1)}$ is upper triangular with g_0 on the main diagonal.
- Hence it has full column rank for any nonzero g_0 .
- By Theorem 1, $\mathbf{A}_{\mathcal{I}}(\mathbf{z}_1, \dots, \mathbf{z}_L) \in \mathbb{C}^{(L+1) \times L}$ has full column rank for every set of distinct generators $\{\mathbf{z}_1, \dots, \mathbf{z}_L\}$.

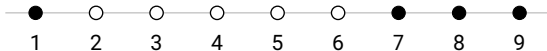
MAXIMUM-CONSECUTIVE-LAG ARRAYS

Corollary 4 in pre-print <http://arxiv.org/abs/2606.00360>



MAXIMUM CONSECUTIVE LAG EXAMPLE

Example 9 of pre-print <http://arxiv.org/abs/2606.00360>

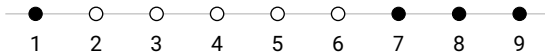


- For $M = 9$, $M_{\mathcal{I}} = 4$ and $L = 3$ sources, the Toeplitz matrix is

$$\mathcal{T}(\mathbf{g}) = \begin{bmatrix} g_0 & g_1 & g_1^* & g_0^* & 0 & 0 & 0 & 0 & 0 \\ 0 & g_0 & g_1 & g_1^* & g_0^* & 0 & 0 & 0 & 0 \\ 0 & 0 & g_0 & g_1 & g_1^* & g_0^* & 0 & 0 & 0 \\ 0 & 0 & 0 & g_0 & g_1 & g_1^* & g_0^* & 0 & 0 \\ 0 & 0 & 0 & 0 & g_0 & g_1 & g_1^* & g_0^* & 0 \\ 0 & 0 & 0 & 0 & 0 & g_0 & g_1 & g_1^* & g_0^* \end{bmatrix}.$$

MAXIMUM CONSECUTIVE LAG EXAMPLE

Example 9 of pre-print <http://arxiv.org/abs/2606.00360>



- The corresponding **complementary submatrix** is

$$\mathcal{T}_{\bar{\mathcal{I}}}(\mathbf{g}) = \begin{bmatrix} g_1 & g_1^* & g_0^* & 0 & 0 \\ \textcolor{red}{g}_0 & g_1 & g_1^* & g_0^* & 0 \\ 0 & \textcolor{red}{g}_0 & g_1 & g_1^* & g_0^* \\ 0 & 0 & \textcolor{red}{g}_0 & g_1 & g_1^* \\ 0 & 0 & 0 & \textcolor{red}{g}_0 & g_1 \\ 0 & 0 & 0 & 0 & \textcolor{red}{g}_0 \end{bmatrix}.$$

- The 5×5 submatrix formed by the last five rows of $\mathcal{T}_{\bar{\mathcal{I}}}(\mathbf{g})$ is **upper triangular** with diagonal entries $\textcolor{blue}{g}_0 \neq 0$, so its determinant is $g_0^5 \neq 0$.
- Hence $\mathcal{T}_{\bar{\mathcal{I}}}(\mathbf{g})$ has **full column rank** for all $\mathbf{g}$, and by Theorem 1 the **array manifold is unambiguous for $L = 3$** .

Array analysis

Alternative rank criterion with fixed generators

FIXING MULTIPLE GENERATORS ON THE UNIT-CIRCLE

- We have previously discussed that any generator $z_1, \dots, z_L$ set can always be rotated by a fixed electrical angle α , i.e., by multiplying all generators with $e^{j\alpha}$.
- This rotation property can, e.g., be used to fix one of the generators at a desired angle, i.e., to ensure for example $z_1 = 1$.
- The question that arises in this context is how **multiple generators** can be **fixed at predefined positions on the unit-circle** in the ambiguity analysis (e.g., L' generators $z_{i_1}, \dots, z_{i_{L'}}$ out of a total of L generators).
- The following results provides an answer to this question, which turns out to be very useful in practice.

VANDERMONDE REDUCTION

Corollary 8 of pre-print <http://arxiv.org/abs/2606.00360>

Corollary 8

Let $\{\acute{z}_1, \dots, \acute{z}_{L'}\} \subset \{z_1, \dots, z_L\}$ be an arbitrary subset of $L' < L$ generators of $P_g(z)$, and let $\mathbf{f}$ be the coefficient vector of the cofactor polynomial $P_f(z)$, so that $P_g(z) = P_f(z) P_h(z)$ with $P_h(z) = \prod_{\ell=1}^{L'} (z - \acute{z}_\ell)$. Then the generalized Vandermonde matrix $\mathbf{A}_{\mathcal{I}}(z_1, \dots, z_L)$ is rank deficient if and only if the $(M - \bar{L}') \times (M_{\bar{\mathcal{I}}} + L')$ augmented matrix

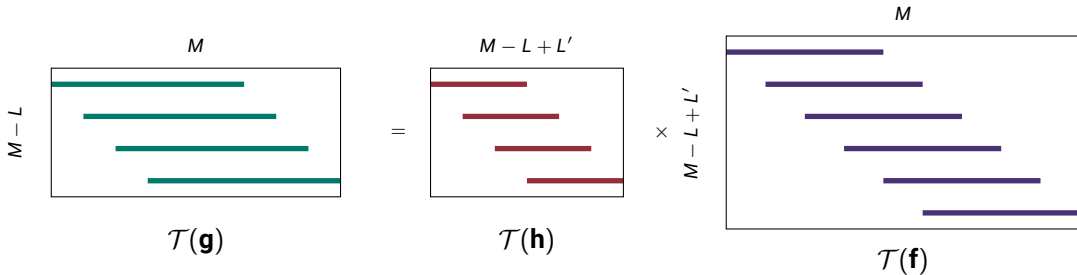
$$\mathcal{A}(\mathbf{f}, \acute{z}_1, \dots, \acute{z}_{L'}) = [\mathcal{T}_{\bar{\mathcal{I}}}(\mathbf{f}) \quad \mathbf{A}_{M-\bar{L}'}(\acute{z}_1, \dots, \acute{z}_{L'})]$$

is rank deficient.

- Corollary 8 relates rank of the **generalized Vandermonde** matrix $\mathbf{A}_{\mathcal{I}}(z_1, \dots, z_L)$ of the thinned array and that of the **full Vandermonde** matrix $\mathbf{A}_{M-\bar{L}'}(\acute{z}_1, \dots, \acute{z}_{L'})$ of a ULA with $M - \bar{L}'$ elements, for any arbitrary subset of L' generators.
- Characterizing ambiguities in the thinned array reduces to characterizing rank deficiencies of an **augmented full-ULA steering matrix** or **extended complementary submatrix**.

MATRIX REPRESENTATION OF POLYNOMIAL MULTIPLICATION

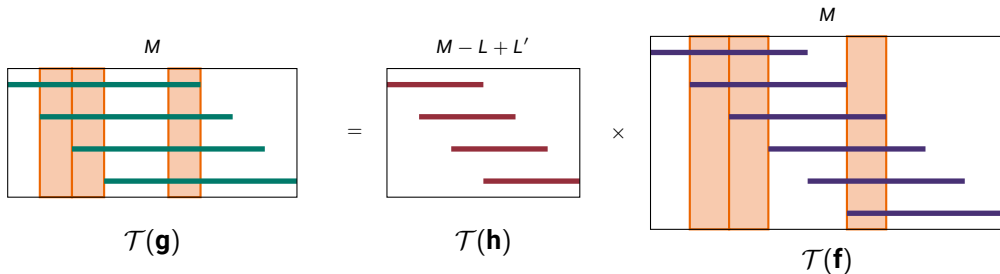
Sketch of proof of Corollary 8. For details see: <http://arxiv.org/abs/2606.00360>



- Prescribing L' of the L generators factors the polynomial: $P_{\mathbf{g}}(z) = P_{\mathbf{h}}(z) P_{\mathbf{f}}(z)$
- This lifts to the matrices: $\mathcal{T}(\mathbf{g}) = \mathcal{T}(\mathbf{h}) \mathcal{T}(\mathbf{f})$, both spanning all M columns

LOCATE $\bar{\mathcal{I}}$ IN BOTH FACTORS

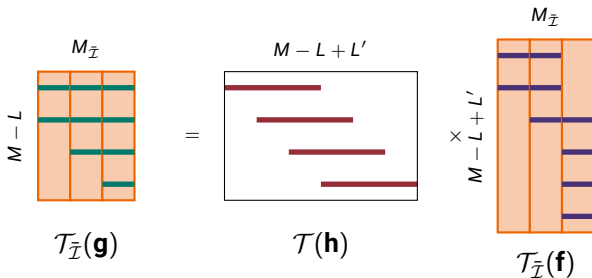
Sketch of proof of Corollary 8. For details see: <http://arxiv.org/abs/2606.00360>



$\mathcal{T}(\mathbf{g})$ and $\mathcal{T}(\mathbf{f})$ share the same M columns and $\bar{\mathcal{I}}$ removal positions

EXTRACTING THOSE COLUMNS

Sketch of proof of Corollary 8. For details see: <http://arxiv.org/abs/2606.00360>



$\mathcal{T}(\mathbf{g})$ and $\mathcal{T}(\mathbf{f})$ share the same M columns and $\bar{I}$ removal positions

THE NULL SPACE OF $\mathcal{T}(\mathbf{h})$

Sketch of proof of Corollary 8. For details see: <http://arxiv.org/abs/2606.00360>

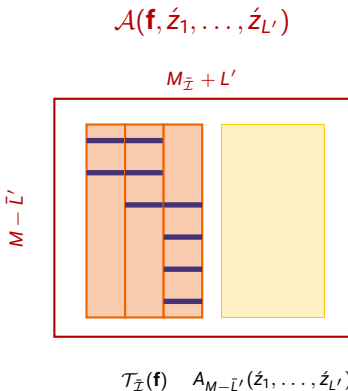
$$\begin{array}{c}
 \begin{array}{c} \overbrace{M-L+L'}^{-L'} \\ \mathcal{T}(\mathbf{h}) \end{array}
 \quad \times \quad
 \begin{array}{c} L' \\ A_{M-\bar{L}'}(\hat{z}_1, \dots, \hat{z}_{L'}) \end{array}
 =
 \begin{array}{c} L' \\ \mathbf{0} \end{array}
 \end{array}$$

Diagram illustrating the null space of $\mathcal{T}(\mathbf{h})$. The matrix $\mathcal{T}(\mathbf{h})$ has dimensions $(M-L) \times (M-L+L')$. The matrix $A_{M-\bar{L}'}(\hat{z}_1, \dots, \hat{z}_{L'})$ has dimensions $(M-\bar{L}') \times L'$. The product of these two matrices is the zero matrix $\mathbf{0}$ of dimensions $L' \times L'$.

Because $\mathcal{T}(\mathbf{h})$ is built from the L' prescribed generators, its null space is exactly the Vandermonde matrix of those generators, with $M - \bar{L}' = M - L + L'$ rows matching $\mathcal{T}(\mathbf{h})$'s column count

THE AUGMENTED MATRIX

Sketch of proof of Corollary 8. For details see: <http://arxiv.org/abs/2606.00360>



$$\mathcal{T}_{\bar{\mathcal{I}}}(\mathbf{f}) \in \mathcal{R}(\mathbf{A}_{M-\bar{\mathcal{I}}'}(\dot{z}_1, \dots, \dot{z}_{L'}))$$



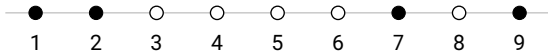
$$\mathcal{A}(\mathbf{f}, \dot{z}_1, \dots, \dot{z}_{L'}) \text{ rank deficient}$$



$$\mathbf{A}_{\mathcal{I}}(z_1, \dots, z_L) \text{ is rank deficient}$$

SPARSE ARRAY WITH LOW MUTUAL COUPLING

Example 10 of pre-print <http://arxiv.org/abs/2606.00360>



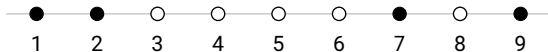
- As an **alternative to Example 9**, we seek a sparse design with $M_{\mathcal{I}} = 4$ sensors and $L = 3$ sources that **avoids the dense contiguous lag** three-element subarray.
- For this array the complementary submatrix is

$$\mathcal{T}_{\bar{\mathcal{I}}}(\mathbf{g}) = \begin{bmatrix} g_1^* & g_0^* & 0 & 0 & 0 \\ g_1 & g_1^* & g_0^* & 0 & 0 \\ g_0 & g_1 & g_1^* & g_0^* & 0 \\ 0 & g_0 & g_1 & g_1^* & 0 \\ 0 & 0 & g_0 & g_1 & g_0^* \\ 0 & 0 & 0 & g_0 & g_1^* \end{bmatrix}.$$

- Rather than directly analyzing the rank of $\mathcal{T}_{\bar{\mathcal{I}}}(\mathbf{g})$, we apply **Corollary 8** with **one prescribed generator $\hat{z}_1 = 1$** , i.e $L' = 1$.
- The **cofactor polynomial** has degree $L - L' = 2$ with coefficient vector $\mathbf{f} = [f_0, f_1, f_0^*]^T$.

SPARSE ARRAY WITH LOW MUTUAL COUPLING

Example 10 of pre-print <http://arxiv.org/abs/2606.00360>



- The associated 7×6 augmented complementary submatrix reads

$$\mathcal{A}(\mathbf{f}, \dot{\mathbf{z}}_1) = \begin{bmatrix} f_0^* & 0 & 0 & 0 & 0 & 1 \\ f_1 & f_0^* & 0 & 0 & 0 & 1 \\ f_0 & f_1 & f_0^* & 0 & 0 & 1 \\ 0 & f_0 & f_1 & f_0^* & 0 & 1 \\ 0 & 0 & f_0 & f_1 & 0 & 1 \\ 0 & 0 & 0 & f_0 & f_0^* & 1 \\ 0 & 0 & 0 & 0 & f_1 & 1 \end{bmatrix}.$$

- It can be shown that the **augmented matrix** $\mathcal{A}(\mathbf{f}, \dot{\mathbf{z}}_1)$ has **full column rank**, i.e., $\text{rank}(\mathcal{A}(\mathbf{f}, \dot{\mathbf{z}}_1)) = 6$ for all $|f_0| = 1$ and $f_1 \in \mathbb{R}$.
- It follows that the associated **generalized Vandermonde** matrix $\mathbf{A}_{\mathcal{I}}(z_1, z_2, z_3)$ is **always full rank**.

SETUP: LINEAR INDEPENDENCE OF THE FIRST FIVE COLUMNS

Example 10 of pre-print <http://arxiv.org/abs/2606.00360>

$$\mathcal{A}(\mathbf{f}, \mathbf{z}_1) = \begin{bmatrix} f_0^* & 0 & 0 & 0 & 0 & 1 \\ f_1 & f_0^* & 0 & 0 & 0 & 1 \\ f_0 & f_1 & f_0^* & 0 & 0 & 1 \\ 0 & f_0 & f_1 & f_0^* & 0 & 1 \\ 0 & 0 & f_0 & f_1 & 0 & 1 \\ 0 & 0 & 0 & f_0 & f_0^* & 1 \\ 0 & 0 & 0 & 0 & f_1 & 1 \end{bmatrix}.$$

Since $f_0 = e^{j\phi_0} \neq 0$, the first five columns of $\mathcal{A}(\mathbf{f}, \mathbf{z}_1)$ are linearly independent:

- the first column has f_0^* in row 1 as a pivot;
- the others can be reduced to staircase form by elementary operations using f_0 on the subdiagonals.

Hence rank deficiency can occur only if the sixth column (the all-ones Vandermonde column for $\mathbf{z}_1 = 1$) lies in the span of the first five.

SETTING UP THE LINEAR SYSTEM

Example 10 of pre-print <http://arxiv.org/abs/2606.00360>

Suppose, for some linear coefficients $\alpha_1, \dots, \alpha_5 \in \mathbb{C}$,

$$\sum_{j=1}^5 \alpha_j [\mathcal{A}(\mathbf{f}, \dot{\mathbf{z}}_1)]_{:,j} = [\mathcal{A}(\mathbf{f}, \dot{\mathbf{z}}_1)]_{:,6} = \mathbf{1}.$$

Writing out the seven row equations gives:

$$\begin{aligned} f_0^* \alpha_1 &= 1, & f_0 \alpha_3 + f_1 \alpha_4 &= 1, \\ f_1 \alpha_1 + f_0^* \alpha_2 &= 1, & f_0 \alpha_4 + f_0^* \alpha_5 &= 1, \\ f_0 \alpha_1 + f_1 \alpha_2 + f_0^* \alpha_3 &= 1, & f_1 \alpha_5 &= 1. \\ f_0 \alpha_2 + f_1 \alpha_3 + f_0^* \alpha_4 &= 1, \end{aligned}$$

The first, second, third, fourth, and sixth equations can be solved *recursively* for $\alpha_1, \dots, \alpha_5$ in terms of f_0 and f_1 . Substituting into the two remaining equations (the fifth and seventh) yields two consistency conditions.

CONSISTENCY CONDITIONS

Example 10 of pre-print <http://arxiv.org/abs/2606.00360>

Substituting the recursively-solved α_j into the remaining two equations yields

$$F_5(f_0, f_1) = 0, \quad F_7(f_0, f_1) = 0,$$

where

$$\begin{aligned} F_5(f_0, f_1) = & -f_0^4 f_1^4 + f_0^3 f_1^3 + (3f_0^4 - f_0^2) f_1^2 \\ & + (-2f_0^3 + f_0) f_1 + (f_0^2 - f_0^4 - 1), \end{aligned}$$

$$\begin{aligned} F_7(f_0, f_1) = & f_0^6 f_1^4 - f_0^5 f_1^3 + (-2f_0^6 + f_0^4) f_1^2 \\ & + (f_0^5 - f_0^3 + f_0) f_1 - 1. \end{aligned}$$

Goal: show these have no common solution with $|f_0| = 1, f_1 \in \mathbb{R}$ (proof by contradiction).

PROOF BY CONTRADICTION

Example 10 of pre-print <http://arxiv.org/abs/2606.00360>

Define

$$F(f_0, f_1) = F_7(f_0, f_1) + f_0^2 F_5(f_0, f_1).$$

The f_1^4 and f_1^3 terms cancel, and direct simplification gives

$$F(f_0, f_1) = f_0^6 f_1^2 - f_0^6 - f_0^5 f_1 + f_0^4 - f_0^2 + f_0 f_1 - 1.$$

TRIGONOMETRIC FORM

Example 10 of pre-print <http://arxiv.org/abs/2606.00360>

Writing $f_0 = e^{j\phi_0}$ and dividing $F(f_0, f_1) = 0$ by $f_0^3 = e^{j3\phi_0}$:

$$\begin{aligned}(f_1^2 - 1) e^{j3\phi_0} - f_1 e^{j2\phi_0} + e^{j\phi_0} - e^{-j\phi_0} \\ + f_1 e^{-j2\phi_0} - e^{-j3\phi_0} = 0.\end{aligned}$$

Separating real and imaginary parts:

$$(f_1^2 - 2) \cos(3\phi_0) = 0, \tag{1}$$

$$f_1^2 \sin(3\phi_0) - 2f_1 \sin(2\phi_0) + 2 \sin(\phi_0) = 0. \tag{2}$$

CASE 1: $\cos(3\phi_0) \neq 0$

Example 10 of pre-print <http://arxiv.org/abs/2606.00360>

Equation (1) forces $f_1^2 = 2$. Substituting into (2) gives

$$\sin(3\phi_0) - f_1 \sin(2\phi_0) + \sin(\phi_0) = 0.$$

Using $\sin(3\phi_0) + \sin(\phi_0) = 2 \sin(2\phi_0) \cos(\phi_0)$, this reduces to

$$\sin(2\phi_0) (2 \cos(\phi_0) - f_1) = 0,$$

giving two sub-cases.

(a) $\sin(2\phi_0) = 0$

$\phi_0 \in \{0, \pi\}$ ($\pm\pi/2$ excluded), so $f_0 = \pm 1$,
 $f_1 = \pm\sqrt{2}$.

$$F_5(\pm 1, \pm\sqrt{2}) \neq 0.$$

(b) $f_1 = 2 \cos(\phi_0)$

With $f_1^2 = 2$: $\cos(\phi_0) = \pm \frac{1}{\sqrt{2}}$, so
 $\phi_0 \in \{\pm\pi/4, \pm 3\pi/4\}$.

$$F_5(f_0, f_1) \neq 0.$$

Both sub-cases contradict the required consistency conditions.

CASE 2: $\cos(3\phi_0) = 0$

Example 10 of pre-print <http://arxiv.org/abs/2606.00360>

Then

$$\phi_0 \in \left\{ \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6} \right\}.$$

Substituting each value into (2) yields either:

- the quadratic $f_1^2 \pm \sqrt{3}f_1 + 1 = 0$, which has *no real solution* (discriminant $3 - 4 = -1 < 0$), or
- $f_0 = \pm j$ together with $f_1^2 = 2$.

In the latter case, direct substitution shows $F_5(\pm j, \pm\sqrt{2}) \neq 0$, again a contradiction.

CONCLUSION

In every case, assuming $F_5(f_0, f_1) = F_7(f_0, f_1) = 0$ with $|f_0| = 1, f_1 \in \mathbb{R}$ leads to a contradiction.

Thus $F_5(f_0, f_1)$ and $F_7(f_0, f_1)$ cannot vanish simultaneously for any $|f_0| = 1$ and $f_1 \in \mathbb{R}$.

Conclusion: the sixth column of $\mathcal{A}(\mathbf{f}, \mathbf{z}_1)$ never lies in the span of the first five columns, and

$$\text{rank}(\mathcal{A}(\mathbf{f}, \mathbf{z}_1)) = 6$$

for all admissible f_0 and f_1 .

Sparse array design

Coarrays

SPARSE ARRAYS: COVARIANCE & RECOVERY

- Signal model: $\mathbf{y} = \mathbf{A}(z_1, \dots, z_L) \mathbf{x} + \text{noise}$
- Observation:** Covariance matrix holds **lags** that are "invisible" to the steering vector!

 $\mathbf{R}_{\text{ULA}}$

$$\begin{bmatrix} 1 & z^{-1} & z^{-2} & z^{-3} & z^{-4} \\ z^1 & 1 & z^{-1} & z^{-2} & z^{-3} \\ z^2 & z^1 & 1 & z^{-1} & z^{-2} \\ z^3 & z^2 & z^1 & 1 & z^{-1} \\ z^4 & z^3 & z^2 & z^1 & 1 \end{bmatrix}$$

thinned $\rightarrow$

 $\mathbf{R}_{\text{sparse}}$

$$\begin{bmatrix} 1 & z^{-1} & z^{-2} & & z^{-4} \\ z^1 & 1 & z^{-1} & & z^{-3} \\ z^2 & z^1 & 1 & & z^{-2} \\ & & & & \\ z^4 & z^3 & z^2 & & 1 \end{bmatrix}$$

- $\mathbf{R} = \mathbb{E} \{ \mathbf{y} \mathbf{y}^H \} = \mathbf{a} \mathbf{R}_{xx} \mathbf{a}^H$ is the covariance matrix of the noiseless measurements
- $\mathbf{R} = \mathbf{a} \mathbf{a}^H$ shown here for a single source

SPARSE ARRAYS: DIFFERENCE CO-ARRAYS (DCA)

- Extracting the entries via an aggregation matrix $\mathbf{J}$ defines the **Difference Co-Array (DCA)**

$$\mathbf{a}_{\text{Co-Array}} = \mathbf{J} \text{vec}(\mathbf{R}_{\text{sparse}})$$

Physical Array

Aperture: D
Sensors: M

Difference Co-Array

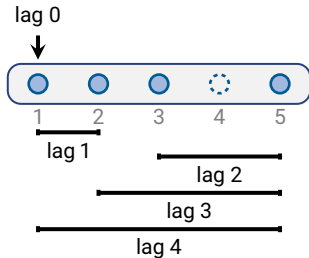
Aperture: $2D - 1$
Sensors: $M_{\text{Co-Array}} > M$

- Larger virtual aperture
- More virtual sensors

⇒ Increased number of resolvable sources

⇒ Question: How many sources can be resolved unambiguously?

SPARSE ARRAYS: DIFFERENCE CO-ARRAYS (DCA)



List of unique lags:
 $\{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$

$\mathbf{R}_{\text{sparse}}$

$$\begin{bmatrix} 1 & z^{-1} & z^{-2} & z^{-4} \\ z^1 & 1 & z^{-1} & z^{-3} \\ z^2 & z^1 & 1 & z^{-2} \\ z^4 & z^3 & z^2 & 1 \end{bmatrix}$$



$\mathbf{a}_{\text{Co-Array}}$

$$\begin{bmatrix} z^{-4} & z^{-3} & z^{-2} & z^{-1} & 1 & z^1 & z^{-2} & z^3 & z^4 \end{bmatrix}^T$$

SPARSE ARRAYS: CO-ARRAY SIGNAL MODEL

- Suppose L **uncorrelated** sources such that $\mathbf{R}_{\mathbf{xx}} = \text{diag}(\mathbf{p})$, $\mathbf{p} = [P_1 \ P_2 \ \dots \ P_L]^T$
- Including **ZMCSCG** noise with covariance $\mathbf{R}_{\mathbf{nn}} = \sigma_n^2 \mathbf{I}$, the overall covariance can be written as

$$\mathbf{R}_{\mathbf{xx}} = \mathbf{A} \mathbf{R}_{\mathbf{xx}} \mathbf{A}^H + \sigma_n^2 \mathbf{I} = \sum_{\ell=1}^L P_{\ell} \mathbf{a}(z_{\ell}) \mathbf{a}^H(z_{\ell}) + \sigma_n^2 \mathbf{I}$$

- Applying the $\text{vec}(\mathbf{R}_{\mathbf{xx}})$ operator yields and the selection operator $\mathbf{J}$ ^[14]

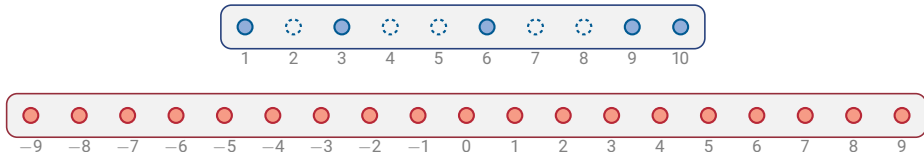
$$\begin{aligned} \mathbf{J} \text{vec}(\mathbf{R}_{\mathbf{xx}}) &= \sum_{\ell=1}^L P_{\ell} \mathbf{J} (\mathbf{a}^*(z_{\ell}) \otimes \mathbf{a}(z_{\ell})) + \sigma_n^2 \mathbf{J} \text{vec}(\mathbf{I}) \\ &= \mathbf{A}_{\text{Co-Array}} \mathbf{p} + \sigma_n^2 \mathbf{J} \text{vec}(\mathbf{I}) \end{aligned}$$

- $\text{vec}(\mathbf{R}_{\mathbf{xx}})$ behaves as a received signal vector with virtual array steering matrix $\mathbf{A}_{\text{Co-Array}}$
- Source signals are replaced by their powers and **noise is deterministic**

[14] M. G. Amin, "Sparse Arrays for Radar, Sonar, and Communications", 2024

SPARSE ARRAYS: MINIMUM REDUNDANCY ARRAY (MRA)

- **Idea:** Minimize the number of physical sensors (i.e., redundant lags) required to obtain a **contiguous co-array** of a given aperture^{[16][3]}
- **Co-Array:** Hole-free DCA: Every lag up to the maximum is present
- **Construction:** No closed-form solution, sensor positions found by **combinatorial search** while preserving the **capability to generate all lags**
- **Drawback:** Computationally expensive to find, does not scale well to large M

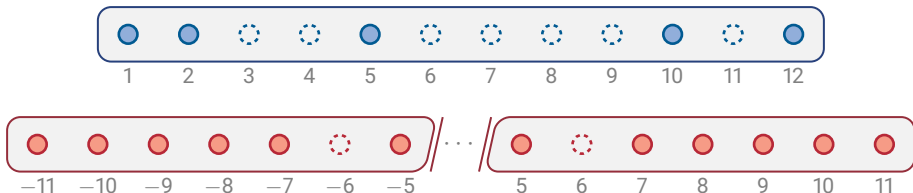


[16] A. Moffet, "Minimum-redundancy linear arrays", 1968

[3] P. Pal and P. P. Vaidyanathan, "Nested Arrays: A Novel Approach to Array Processing With Enhanced Degrees of Freedom", 2010

SPARSE ARRAYS: MINIMUM HOLE ARRAY (MHA) / GOLOMB RULER

- **Idea:** For a fixed number of sensors M , **minimize holes** in the difference co-array^{[17][15]}
- **Co-Array:** Few isolated missing lags, but every lag **only appears once**
- **Construction:** Also search-based, no general closed-form expression
- **Relation to MRA:** Maximum contiguous lag (MRA) vs. minimal occurrence of lags (MHA)
- **Combination** of MRA and MHA is called **perfect**, but only exists for $M \leq 4$

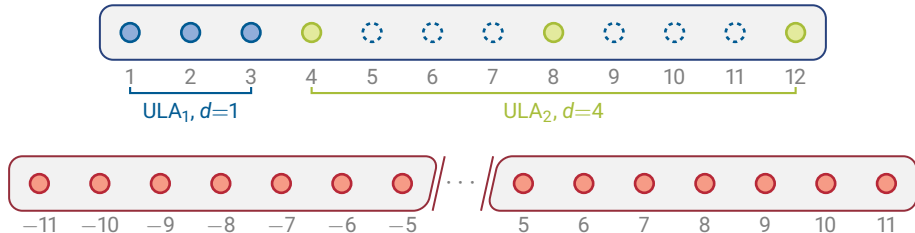


[17] G. Bloom and S. Golomb, "Applications of numbered undirected graphs", 1977

[15] X. Li et al., "Sparse Linear Arrays for Direction-of-Arrival Estimation: A Tutorial Overview", 2025

SPARSE ARRAYS: NESTED ARRAY (NA)

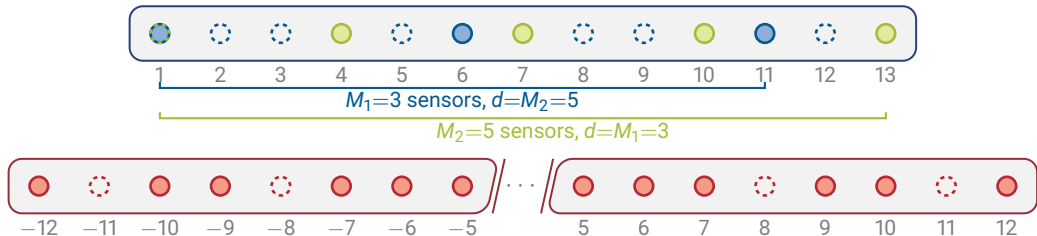
- **Idea:** Concatenate two ULA sub-arrays: **Dense inner** ULA of M_1 sensors with spacing d and **sparse outer** ULA of M_2 sensors with spacing $(M_1 + 1)d^{[3]}$
- **Co-Array:** Contiguous DCA with $O(M_1 M_2)$ virtual sensors
- **Construction:** **Closed-form:** sensor positions known analytically
- **Drawback:** Some lag redundancy, mutual coupling between dense inner sensors



[3] P. Pal and P. P. Vaidyanathan, "Nested Arrays: A Novel Approach to Array Processing With Enhanced Degrees of Freedom", 2010

SPARSE ARRAYS: COPRIME ARRAY (CA)

- **Idea:** Two ULA sub-arrays with **coprime inter-sensor spacings** M_1 and M_2 ($\gcd(M_1, M_2) = 1$), with M_1 and M_2 sensors respectively^[4]
- **Co-Array:** $O(M_1 M_2)$ virtual sensors, large contiguous region
- **Construction:** **Closed-form:** sensor positions given directly by M_1, M_2
- **Advantage:** Reduced mutual coupling compared to NA; flexible design via choice of M_1, M_2



[4] P. P. Vaidyanathan and P. Pal, "Sparse Sensing With Co-Prime Samplers and Arrays", 2011

IMPORTANT CRITERIA FOR SPARSE ARRAY DESIGN

- Optimized **beampattern**^[15]
 - Control mainlobe width and / or maximize virtual aperture
- Maximized **degrees of freedom (DOF)**^[3]
 - DOF determined by the **difference co-array (DCA)**
 - Number of resolvable sources $\leq$ effective DCA aperture / virtual elements
- Mitigation of **mutual coupling**^[18]
 - Distorts array manifold and degrades estimation accuracy
- Optimized **estimation performance bounds**^[1], e.g., Cramér–Rao Bound (CRB)^[2]
 - Directly linked to achievable parameter estimation accuracy

[15] X. Li et al., “**Sparse Linear Arrays for Direction-of-Arrival Estimation: A Tutorial Overview**”, 2025

[3] P. Pal and P. P. Vaidyanathan, “**Nested Arrays: A Novel Approach to Array Processing With Enhanced Degrees of Freedom**”, 2010

[18] C. Larmour et al., “**Sparse Array Mutual Coupling Reduction**”, 2024

[1] J. Tabrikian et al., “**Cognitive Antenna Selection for Automotive Radar Using Bobrovsky-Zakai Bound**”, 2021

[2] I. van der Werf et al., “**Jointly Optimal Array Geometries and Waveforms in Active Sensing: New Insights Into Array Design via the Cramér-Rao Bound**”, 2025

Sparse array design

Ambiguity-based array design

RETURNING TO THEOREM 1

Insights on sparse array design

An immediate question arises:

Can the rank-drop property of [Theorem 1](#) be used to gain insights into [design](#) of [ambiguity-free sparse arrays](#)?

The answer is affirmative:

- Considering [specific choices](#) of coefficient vectors $\mathbf{g}$, entire columns can become [zero](#), or a [linear dependence](#) between [specific columns](#) of the complementary submatrix $\mathcal{T}_{\bar{\mathcal{I}}}(\mathbf{g})$ can be created.
- The sensors corresponding to zero or redundant columns [cannot jointly be deleted](#).
- The design rules obtained from this consideration largely reduce the search-space of possible [ambiguity-free](#) sparse array configurations.

COROLLARY 5: FORBIDDEN CENTER ELEMENTS

$2L \geq M + 1$: removing **any** element of $Q = \{M-L+1, \dots, L\}$ causes ambiguity.



$$M = 11, \quad L = 8$$

Affected submatrix (all of $Q = \{4, \dots, 8\}$ removed, $\bar{\mathcal{I}} = Q$):

$$\mathcal{T}_{\bar{\mathcal{I}}}(\mathbf{g}) = \begin{bmatrix} g_3 & g_4 & g_3^* & g_2^* & g_1^* \\ g_2 & g_3 & g_4 & g_3^* & g_2^* \\ g_1 & g_2 & g_3 & g_4 & g_3^* \end{bmatrix}$$

Every column of $\mathcal{T}_{\bar{\mathcal{I}}}(\mathbf{g})$ is built entirely from $g_1, \dots, g_4$.

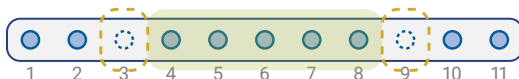
Rank deficient: choose, e.g.,

$$g_0 = g_0^* = 1, \quad g_k = 0 \quad \forall k \neq 0$$

Then $\mathcal{T}_{\bar{\mathcal{I}}}(\tilde{\mathbf{g}}) = \mathbf{0}_{3 \times 5}$, the *entire* submatrix vanishes, so $\text{rank} = 0$ (maximal deficiency).

COROLLARY 6: MUTUALLY EXCLUSIVE INDICES (TYPE A)

$2L \geq M - 1$: removing **both** $M-L$ and $L+1$ simultaneously causes ambiguity.



$$M = 11, \quad L = 8$$

Shaded region (4–8): already forced retained by Corollary 5

Affected submatrix ($\tilde{\mathcal{I}} = \{3, 9\}$):

$$\mathcal{T}_{\tilde{\mathcal{I}}}(\mathbf{g}) = \begin{bmatrix} g_2 & g_0^* \\ g_1 & g_1^* \\ g_0 & g_2^* \end{bmatrix}$$

Rank deficient: choose, e.g.,

$$\mathbf{g} = \mathbf{1} \quad (g_0 = \dots = g_4 = 1)$$

Columns 3, 9 both equal $[1, 1, 1]^T$.

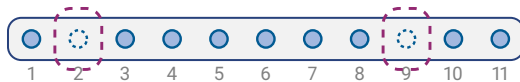
Columns 3 and 9 share the same coefficients in mirrored order.



COROLLARY 7: MUTUALLY EXCLUSIVE INDICES (TYPE B)

$$N = M - L, \quad 0 \leq p \leq K, \quad p + 1 \leq q \leq L - p, \quad N \leq L + q - 2p - 1$$

Removing **both** $S_{p,q} = \{q, L - p + q\}$ simultaneously causes ambiguity.



$$M = 11, \quad L = 8$$

Choosing $p = 1, q = 2$ returns $\bar{\mathcal{I}} = \{2, 9\}$:

$$\mathcal{T}_{\bar{\mathcal{I}}}(\mathbf{g}) = \begin{bmatrix} g_1 & g_0^* \\ g_0 & g_1^* \\ 0 & g_2^* \end{bmatrix}$$

Rank deficient: choose, e.g.,

$$g_0 = g_1 = 1, \quad g_2 = g_4 = 0 \quad g_3 \text{ free}$$

Columns 2, 9 both equal $[1, 1, 0]^T$.

Conclusions and outlook

CONCLUSIONS AND OUTLOOK

We addressed a fundamental problem:

- The **Kruskal rank** (Spark) of **Generalized Vandermonde** matrix
- A **scalable** and **complete characterization** of rank drop conditions
- Results can be used for sparse array **analysis and design**

Future work:

- **Optimal sparse array design**: Find further results
- Sparse **co-array design**
- Sparse **2D-array analysis and design**
- More **general arrays**: Shift invariant arrays, partly calibrated arrays
- Consider particular array cases
- Transfer results to **other applications**: System identification in control,...

Please check the associated pre-print: <http://arxiv.org/abs/2606.00360>

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ANY QUESTIONS ?

Find more details in the pre-print: <http://arxiv.org/abs/2606.00360>

Please contact us if you have ideas and interest for collaborations!



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